

**Near-fault Ground Motion Prediction, Site Effects and
Wave Propagation at Regional Scale for a Shallow
Earthquake in Southeastern France**
**Évaluation du mouvement sismique en champ proche, prise
en compte des effets de site et de propagation régionale
pour un séisme superficiel dans le Sud-est de la France**

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Présentée par :

Hongyi SU

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Directeur de thèse: Mathieu CAUSSE (ISTerre)

Co-encadrant de thèse: Sébastien HOK (ASNR)

Cécile CORNOU (ISTerre)

Céline GÉLIS (ASNR)

Le jury composé de: X, Rapporteurs

X, Rapporteurs

X, Examiner

X, Examiner

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0.1 Overview for Chapter Two

In this chapter, I adopt a completely different approach to achieve the goal of simulating near-fault ground motion. Unlike the dynamic rupture modelling method presented in the first chapter, which is physics-based and prioritizes physical consistency. In contrast, the method used in this chapter is more 'chaotic', the method known as *stochastic kinematic modelling*. Unlike the physics-based dynamic rupture modelling, the observation is the 'king' in this approach.

If I may draw an analogy with Charles Darwin's theory of evolution, its underlying principles are remarkably similar to those of stochastic kinematic source modeling. In this modeling framework, I first generate thousands of random rupture models, a step that corresponds to random variation and overpopulation in evolutionary theory. Second, I filter these models based on how well they fit the field observations using forward modeling and Bayesian updating—a process directly analogous to natural selection. Finally, this yields the posterior distributions for the near-fault ground motion and source parameters. Models that successfully reproduce the observations are rewarded with high statistical weights, while those that fail are penalized with low weights, effectively reducing their "chance of survival." Just as highly adapted individuals become more prevalent in nature while less adapted traits gradually disappear, the most accurate models ultimately dominate the final posterior distribution.

The key findings in this chapter are summarized as follows:

First, I define five input source parameters, their assumed probability distributions and the applied prior knowledge. The input source parameters are nucleation position along dip (ynuc), nucleation position along strike (xnuc), corner frequency, mean rise time, and weak zone taper strength. Figure 1.7 illustrates the populated 1000 data samples and their prior distributions. This set of input source parameters statistically checked by four earthquake source features: (1) the target corner frequency from the source spectra matching the best-fitting Brune model[3]; (2) the rupture duration agrees with an empirical model [8]; (3) the moment rate function, maintaining consistency with the preferred dynamic rupture model in Chapter 1; and (4) the vertical surface displacement aligns with InSAR measurements [30].

The stochastic kinematic source modeling code was originally written by M. Causse. I parallelized the code 'Kinspy' for cluster execution to efficiently compute the ground motion up to 10 Hz and on a velocity model with a shallow layer (minimum $V_s = 547$ m/s at around 20 meters). Moreover, I also implemented a weak zone taper to reduce high-frequency energy radiation from shallow depths, based on studies by [14] and [27]. Furthermore, I introduced a physical constraint ensuring that the local rupture velocity never exceeds the P-wave velocity of the respective layer.

Second, I explain the observational datasets and the forward modeling to predict these observations, which are:

- **Overtured Rocks:** Located on the fault trace. I filtered out all simulations that failed to produce a downward vertical PGA exceeding 1g, and retaining 40.6% of the initial models (Figure 1.18).
- **Graveyard Slab Displacements:** Displacements believed to have been caused by co-seismic ground shaking. The forward modeling utilizes an updated MATLAB code originally written by Laurent Baillet [4], which I modified to randomize the input static friction coefficient (Figure 1.35).
- **Building Damage:** Post-event damage reports from the nearby city of Le Teil [33]. The forward modeling relies on a pretrained Artificial Neural Network (ANN) model [20]. To achieve this, I simulated ground motion at approximately 49 virtual stations near the 306 damaged buildings. These stations were strategically placed to sufficiently interpolate the PGA and PGV values beneath each structure. The resulting ANN predictions for the building damage grades are shown in Figures 1.22 and 1.23.

Finally, I demonstrate how Bayesian updating is used to derive the posterior distributions of both the near-fault ground motion (PGA) and the input source parameters based on these non-instrumental observations:

- **Using building damage data:** I extracted the posterior PGA for 49 stations near the 306 damaged buildings (Figure 1.31). As shown in Figure 1.32, the Bayesian update results in a mean PGA reduction of $52\% \pm 3\%$ across all stations.
- **Using graveyard slab observations:** I performed a Random Forest analysis, which indicates that the input source parameters can be meaningfully constrained by this data, especially the mean rise time. As shown in Figure 1.50, the posterior distribution for the static friction coefficient shows a distinct preference for $\mu = 0.2$ (out of four discrete values: 0.1, 0.2, 0.3, and 0.4). While the posterior distribution for the mean rise time remains broadly similar to the prior, another notable change occurs in the corner frequency, where the highest posterior peak corresponds to a mean rupture velocity of approximately 1500 m/s.

Research Dissemination: An abstract about the work in this chapter has been accepted and will be presented at the 7th IASPEI / IAEE International Symposium on the Effects of Surface Geology on Seismic Motion (ESG7) in Grenoble, France, from October 19–21, 2026.

Chapter 1

Stochastic Kinematic Rupture Modelling

1.1 Summary

In this chapter, I apply a different methodology for generating kinematic source and near-fault ground motions.

First, the kinematic source generation is based on a stochastic method. Unlike dynamic rupture models, this approach does not model friction behavior on the fault plane; instead, the rupture parameters are imposed based on predefined distributions derived from known knowledge. For example, the corner frequency (F_c) is constrained by seismological data [4]. All input source parameters will be detailed in the following section, it is important to emphasize here that the variability of source parameters explored using stochastic approach is significantly larger and efficient than what is typically achieved with dynamic rupture models.

Second, the near-fault ground motions (seismograms) are computed using Axitra code [9, 7]. The seismograms are computed by convolving the source time function of each subfault along a finite fault with corresponding Green's functions. These functions account for the wave propagation effects and the input velocity model between each subfault and the virtual receivers.

From these simulated seismograms, I extract and establish a prior peak ground acceleration (PGA). To constrain this prior distribution, I utilize three sets of non-instrumented observations near the fault: (1) a site where overturned rocks (ORS) were discovered, figure 1.1 shows an example of displaced rocks from [4] (2) a graveyard where rigid slabs was displaced during the earthquake, figure 1.2 showing an example of the movement of graveslab, and (3) building damage in the nearby city of Le Teil, figure 1.3 from [31] showing few damaged buildings.

By simulating ground motions for thousands of kinematic rupture scenarios, I can constrain the resulting prior PGA distributions by comparing them the non-instrumented observations. Through forward modeling, each simulated scenario is tested against three observable effects:

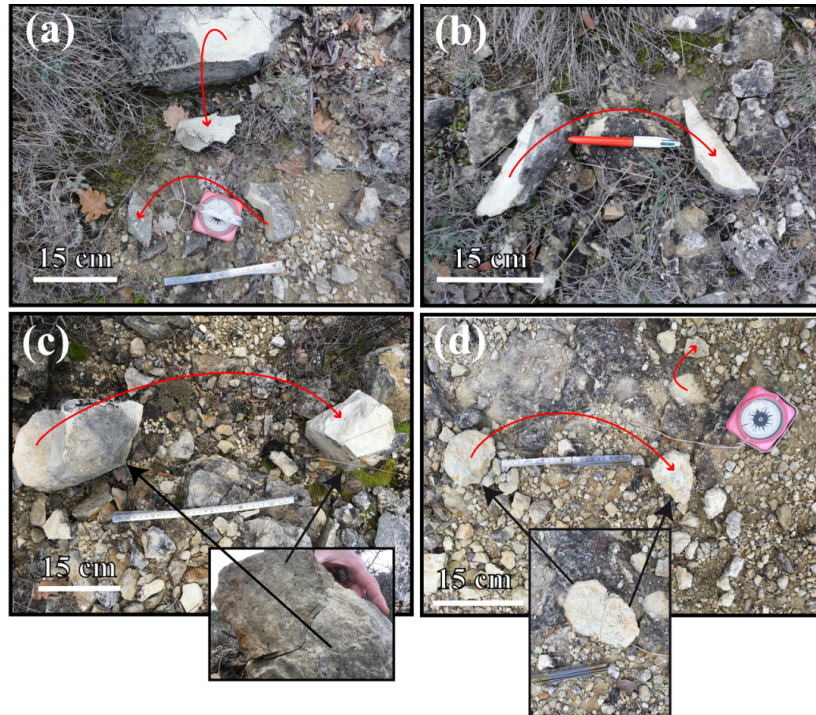


Fig. 1.1 Example of displaced and overturned rocks (ORS) observed near the fault, used as a non-instrumented observation to constrain peak ground acceleration. Photograph from [4].



Fig. 1.2 Displacement of a rigid graveslab observed in a nearby cemetery, providing evidence of strong ground motion during the earthquake. photo credit: Cécile Cornou

- **Overturned rocks:** A virtual station is placed at the rock site. Any rupture model that fails to produce a downward vertical PGA greater than 1g (the force required to overturn the rocks) is rejected.

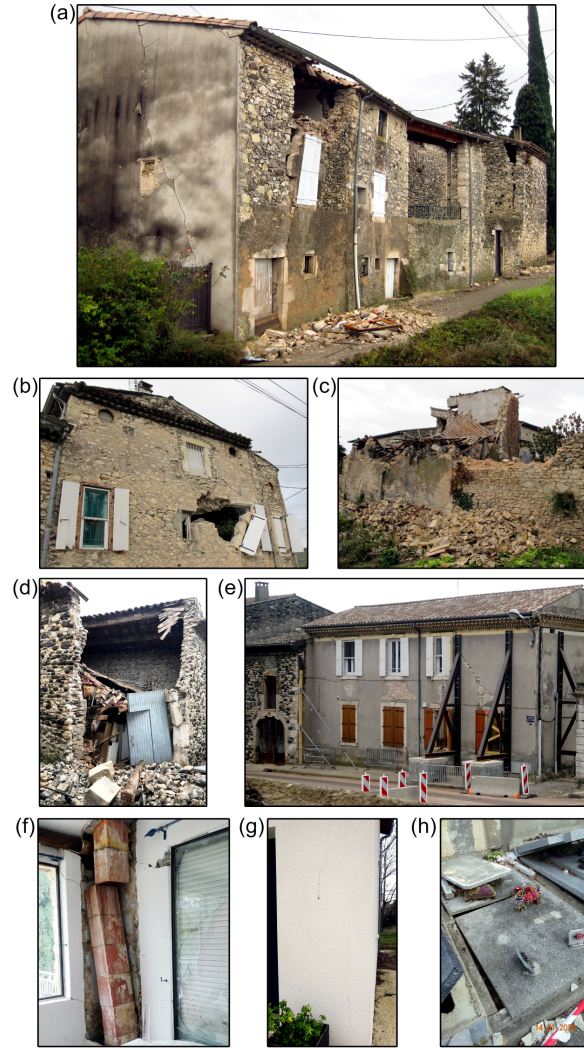


Fig. 1.3 Example of structural damage to a building in the nearby city of Le Teil following the earthquake. Photograph from [31].

- **Displaced graveyard slab:** The simulated 3D ground shaking history (seismogram) at the graveyard site serves as the input for a Coulomb friction model [4] to predict the observed displacement of the slab.
- **Building damage:** 49 virtual stations are distributed across the affected area in Le Teil. The simulated PGA and PGV values are interpolated at under each structure. These values, combined with structural and site-specific parameters, are then fed into a pre-trained Artificial Neural Network (ANN) [20] to predict the resulting damage state.

Figure 1.4 presents the spatial distribution of the total 306 damaged buildings, the proposed virtual receivers situated near these structures. The structural damage data relies on 485 post-earthquake on-site inspections conducted by the French Association of Earthquake Engineering (AFPS) [33]. Of these inspections, 306 buildings were successfully cross-referenced with the French National Building Database (BNDB) to extract metadata regarding building heights and construction dates. The inventory consist of two construction material

classes: masonry and reinforced concrete. The map also show the surface projection of the causative fault alongside localized observation sites: the overturned rock site and the graveyard site (site coordinates: M. Causse, personal communication). Furthermore, the relative contributions of source effects, site effects, and building typology to the observed damage are analyzed in Figure 1.21, revealing that building typology exerts a dominant control on the damage distribution.

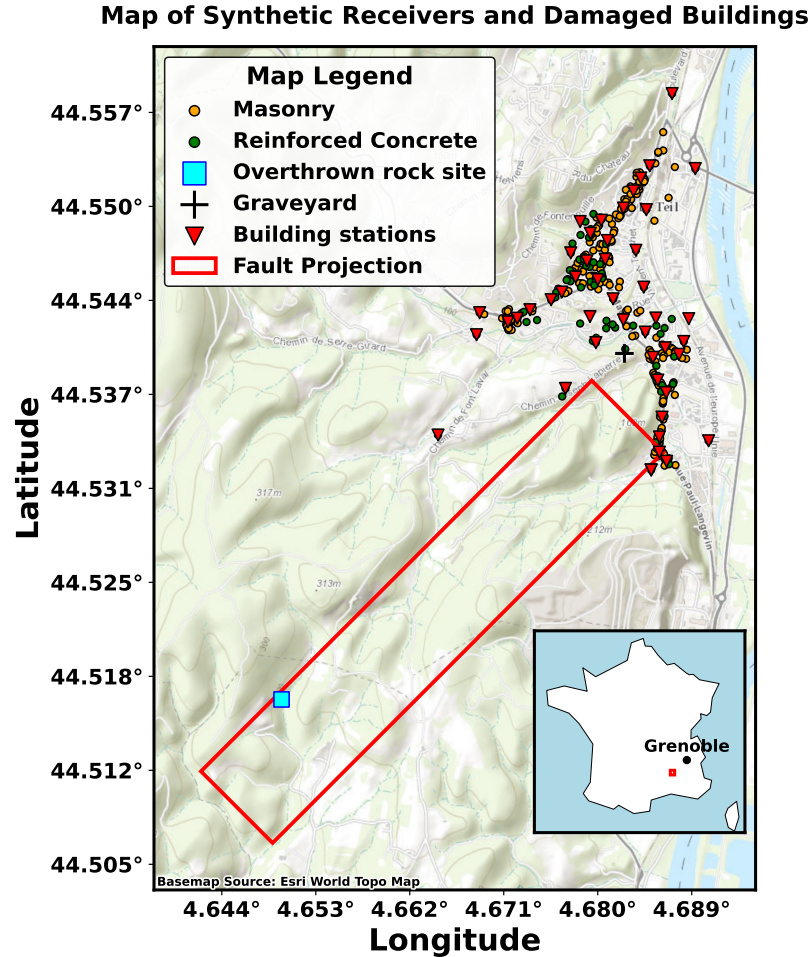


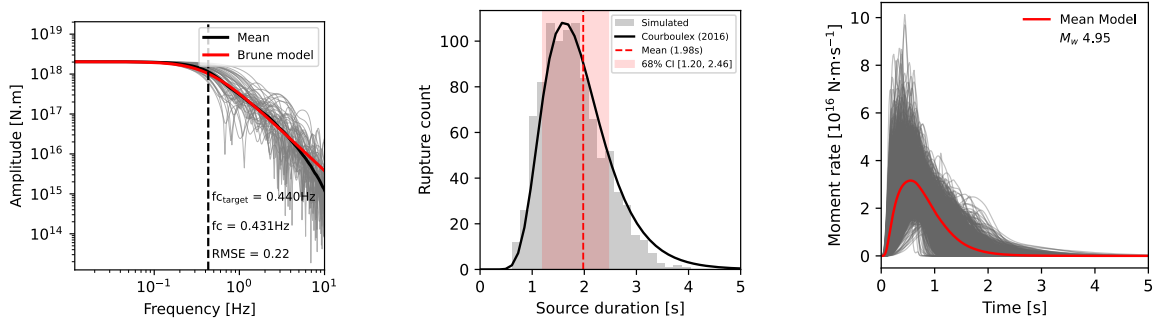
Fig. 1.4 Map of the study area illustrating the spatial distribution of inspected buildings, proposed virtual receivers near the damaged buildings. The map integrates post-earthquake damage assessment data from the AFPS [33]. The map also includes the surface projection of the fault plane, the overturned rock site, and the graveyard site. The damaged buildings are colored by their building topology, which are wither masonry or reinforced concrete

Using these forward models, I perform a Bayesian update on the prior PGA distributions. The likelihood functions are calculated for building damage and graveyard slab displacement observations. For the overturned rock site, this process acts as a simple filter, isolating ruptures that generate downward vertical PGA exceeding 1g.

1.2 Construct Prior PGA

In this section, I detail the computation of the prior peak ground acceleration (PGA), including the definition of each input source parameter and their respective input distributions. I also introduce the input velocity model and analyze the impact of the maximum resolved frequency (cutoff frequency) and the shallow layer structure (weathered rock versus hard rock). Specifically for the Graveyard site, which is located very close to the fault (less than 100 meters), I document the effects of the relative distance to the fault, the relative elevation between the fault and the site, and the impact of being located on the footwall versus the hanging wall.

Before diving into the details, Figure 1.5 provides an overview of the performance of the statistic mean over 1000 stochastic rupture realizations. In the stochastic method, individual ruptures are not meaningful on their own; the focus is on the mean model. Figure 1.5 (a) shows the source spectra (amplitude Fourier spectra of the source time function) for the 1000 rupture realizations (gray lines). The black line represents the mean, while the red line shows the best-fitting Brune (1970) source model [3]. Panel (b) shows the histogram of rupture durations (duration of the moment rate function) compared to a non-subduction empirical model with seismic moment $M_0 = 3.0 \times 10^{16}$ N·m, corresponding to $M_w \approx 4.92$ [8]. Panel (c) shows the moment rate functions for all 1000 realizations (gray) and their mean (red). The mean model has a moment magnitude of $M_w = 4.95$, which is within the acceptable range (68% CI 4.65 - 5.05) [4], and this moment rate function is comparable to the reference dynamic rupture model with the same fault geometry shown in Chapter 1.



(a) Source spectra for 1000 rupture realizations with best-fitting Brune model.

(b) Histogram of rupture durations compared to a non-subduction empirical model.

(c) Moment rate functions of all 1000 realizations and their mean.

Fig. 1.5 Overview of the stochastic rupture performance. Panel (a) shows the amplitude Fourier spectra of 1000 realizations with the mean and best-fitting Brune model; panel (b) shows the rupture duration distribution compared to an empirical non-subduction model; panel (c) shows the moment rate functions for all realizations and their mean.

Moreover, since the Le Teil earthquake has been observed to rupture to the surface and its shallow source nature allows InSAR measurements to map the surface vertical displacement with good resolution [30][6][36], in Figure 1.6 (a) I extracted the slip from the first row of

non-zero subfaults and converted it to vertical displacement using a dipping angle of 57° . The simulated vertical displacement is comparable to both InSAR and field measurements. The sketch in Figure 1.6 (b) illustrates the fault mechanism and the measurable parameters.

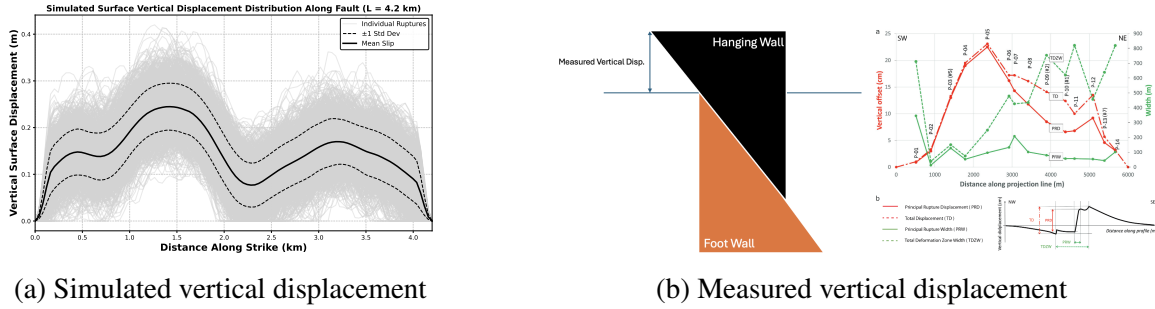


Fig. 1.6 (a) Simulated surface vertical displacement assuming a dip angle of 57° . (b) A schematic diagram illustrating the fault mechanism and the measured parameter, alongside field measurements of surface vertical displacement from [30].

1.2.1 Input Source Parameters

I explore five source parameters: the along-strike (x_{nuc}) and along-dip (y_{nuc}) nucleation positions, the corner frequency (f_c), the mean rise time (τ_{max}), and a weak-zone (WZ) taper strength parameter that controls the reduction of high-frequency energy in the shallow weak zone. Figure 1.7 shows the pairwise relationships among the five source parameters and the mean rupture velocity V_{rup} , which is a function of the input corner frequency and the nucleation position. The mean rupture velocity is calculated as

$$V_{\text{rup}} = \text{clip} \left(\frac{f_c D}{\alpha}, V_{\text{rup,min}}, V_{\text{rup,max}} \right), \quad (1.1)$$

where f_c is the corner frequency, and

$$D = \sqrt{L \cdot x_{\text{nuc}}^2 + W \cdot y_{\text{nuc}}^2}$$

defines a representative rupture length scale, with $L = 4.2$ km representing the fault length and $W = 1.4$ km the fault width. The parameter α is a scaling factor controls the mean rupture velocity and corner frequency. In my thesis, α was set to 0.5; lower values of α result in higher mean rupture velocities and higher estimated corner frequencies. This parameter was tuned to ensure that the estimated corner frequency matched the target value of 0.44 Hz as closely as possible (Figure 1.5a). The parameters $V_{\text{rup,min}}$ and $V_{\text{rup,max}}$ define the lower and upper bounds of the mean rupture velocity, respectively. This parameterization results in a mean rupture velocity of 2.46 ± 0.88 km/s (Table 1.1).

The pair plot, figure 1.7, confirms that all five input parameters are statistically independent, while V_{rup} shows a positive correlation with the corner frequency.

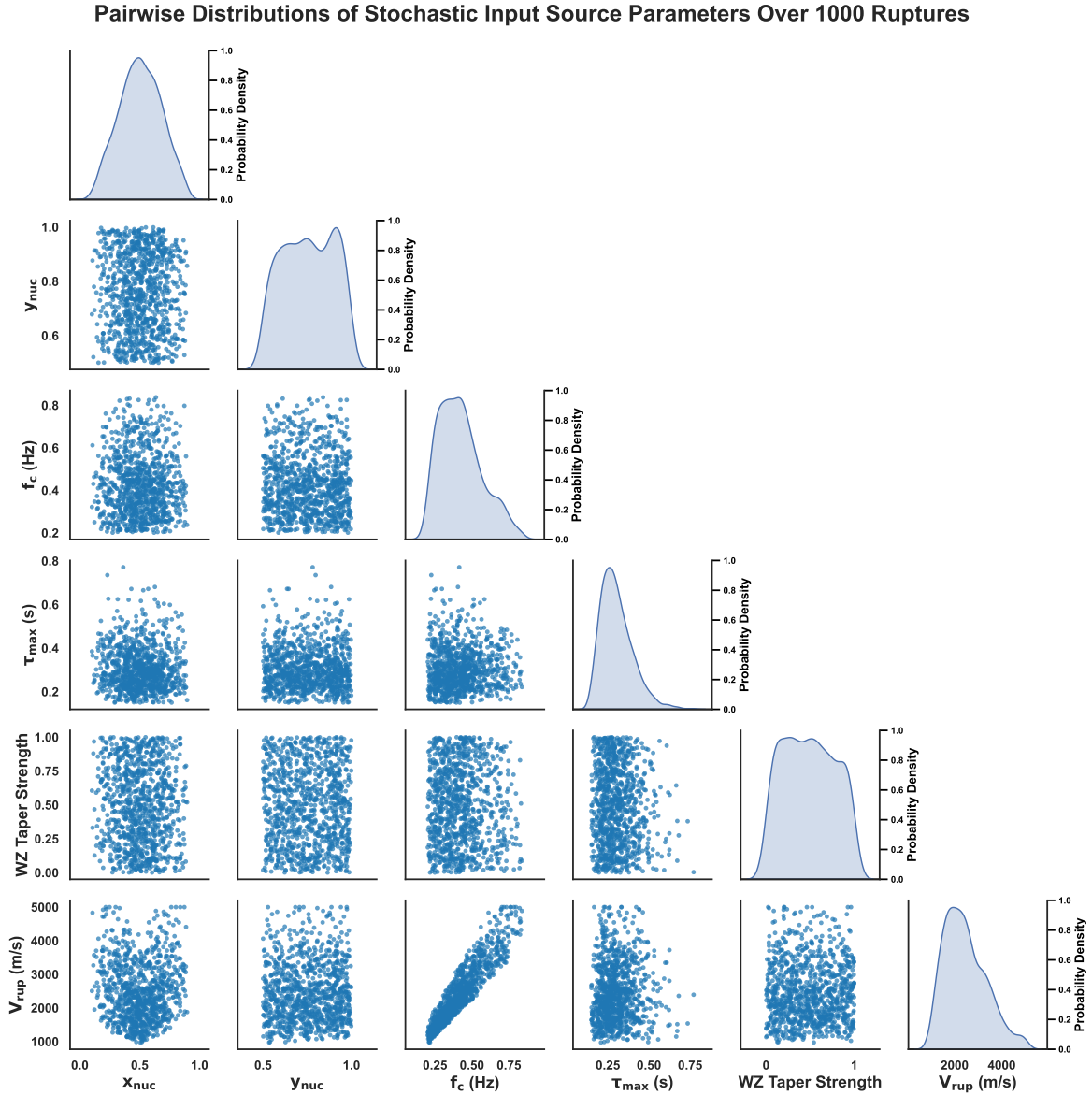


Fig. 1.7 Pair plot of all input parameters. Off-diagonal panels show the relationships between pairs of input parameters, while diagonal panels show the kernel density estimate (KDE) of each parameter.

The presence of a weak zone associated with the Le Teil earthquake has been documented by [30], and similar low-velocity fault zones have been observed worldwide [5, 22, 12, 37, 2]. Weak zones are known to reduce high-frequency radiation, thereby lowering simulated ground-motion amplitudes, while also significantly influencing rupture dynamics [19, 18]. In this study, the approach proposed by [15] and [27] is adopted to incorporate weak-zone effects into the rupture model. Accordingly, a weak-zone strength parameter is introduced to control the degree of tapering within the weak zone. In the present setup, this parameter increases the rise time while reducing the rupture velocity within the weak zone. The impact of weak-zone tapering on the mean rise time and rupture velocity is discussed in detail in Section 1.2.2. Moreover, the shape of the source time function (regularized Yoffe type; [34]) also differs

inside and outside the weak zone. Within the weak zone, the source time function becomes more symmetric and bell-shaped figure 1.14(b), which helps reduce the peak slip rate and consequently lowers high-frequency radiation, this source time function shape difference within weak zone and on hard rock is confirmed by dynamic rupture modeling results [27].

1.2.2 How Seismogram is Computed From Input Source Parameters

First, I impose a low-frequency slip distribution derived from Interferometric SAR (InSAR) data as a starting point. The static slip distribution provide a reference fault-plane slip, as shown in Figure 1.8 [6].

To generate the slip field, a k-square method in the wavenumber domain (ref Methods: Kinematic rupture modeling in [4]) is used to introduce stochastic heterogeneity on top of the low-frequency slip. The resulting slip field captures both the long-wavelength structure from InSAR data and the high-frequency stochastic heterogeneity, a realization shown in figure 1.14a.

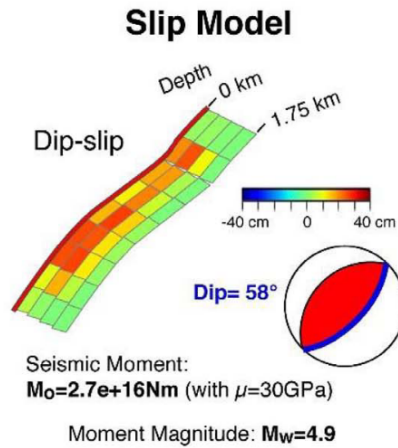


Fig. 1.8 Static slip distribution inverted from the InSAR dataset [6]

From this slip distribution, I generate rise time field with a correlation coefficient of 0.8 with the slip field and a prescribed mean rise time constraint. Then, I apply the WZ taper parameter to locally double the rise time on average within the weak zone to reduce the high energy radiation in shallow depths.

Next, I compute the peak slip rate field using the slip field and rise time field, assuming a Yoffe source time function (STF) [34]. For the shape of the Yoffe STF, I define a more bell-shaped form within the weak zone [27]. The peak slip rate field is local bounded by a maximum value of 4 m/s [4].

I then estimate the rupture velocity field using a correlation coefficient of 0.4 with the peak slip rate field, while also enforcing a mean rupture velocity constraint, which is derived from the input corner frequency and the nucleation position. The mean rupture velocity is bounded between 500 m/s and 5000 m/s, and the local rupture velocity cannot exceed the V_p of the velocity structure.

The correlation parameters between slip field and rise time field, $\text{corr}(\tau, \delta)$, and between peak slip rate (PSR) and rupture velocity (V_r), $\text{corr}(\text{PSR}, V_r)$, are based on statistical analyses of 315 dynamic strike-slip rupture models [32].

The rupture time is subsequently computed from the nucleation position and rupture velocity using the Podvin algorithm [28]. The Yoffe source time function is then combined with the rupture time to construct, for each subfault, a slip-rate time series shifted by its rupture onset time. Finally, this source time function is convolved with precomputed Green's functions from AXITRA for each subfault, and the resulting subfault contributions are summed to generate the final multiple-source seismograms.

Nucleation Position

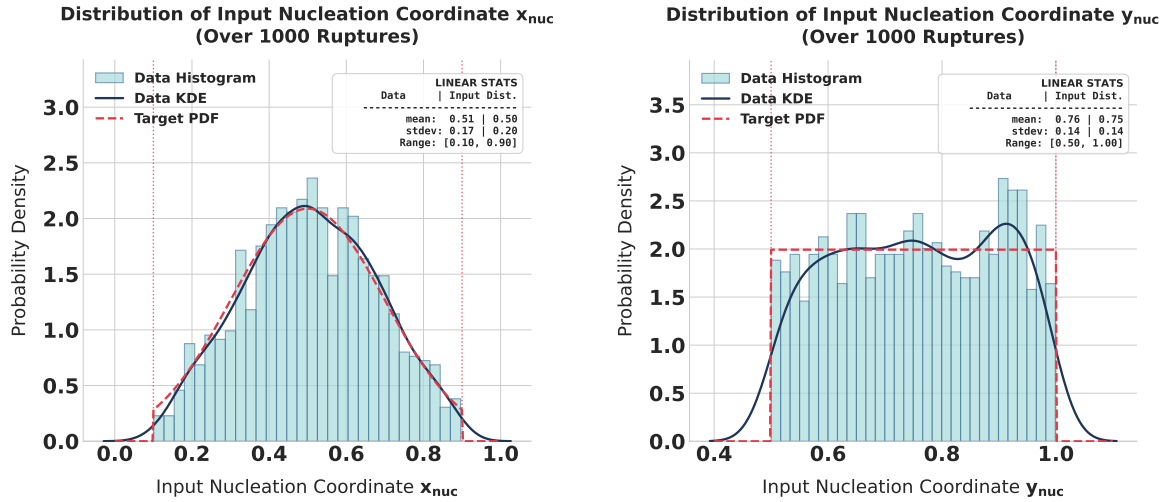
The nucleation position on the fault plane is defined by two parameters, x_{nuc} and y_{nuc} . The parameter x_{nuc} specifies the normalized along-strike position, while y_{nuc} specifies the normalized along-dip position. These parameters are expressed as proportions of the fault length L (4.2 km) and fault width W (1.4 km), respectively. The physical nucleation coordinates are therefore computed as $x_{\text{nuc}} \cdot L$ along strike and $y_{\text{nuc}} \cdot W$ along dip.

The fault plane coordinate system is defined such that the upper-left corner corresponds to $(0, 0)$, the upper-right corner to $(L, 0)$, the lower-left corner to $(0, W)$, and the lower-right corner to (L, W) . For example, $x_{\text{nuc}} = 0.5$ indicates that the rupture nucleates at the midpoint of the fault length, while $y_{\text{nuc}} = 0.5$ indicates nucleation at the midpoint of the fault width.

Figure 1.9 compares the generated samples and prescribed input distributions for x_{nuc} and y_{nuc} . The histograms represent the generated samples, the black curves show the kernel density estimates (KDEs), and the red dashed curves indicate the target probability density functions (PDFs) used as input distributions.

For x_{nuc} , I assume a normal distribution with a mean of 0.5 and a standard deviation of 0.2, favoring nucleation near the center of the fault along strike. To prevent nucleation too close to the fault edges, I apply lower and upper cutoff values of 0.1 and 0.9. Consequently, the prescribed distribution is a truncated normal distribution.

For y_{nuc} , I assume a uniform distribution between 0.5 and 1, restricting nucleation to the lower half of the fault plane. This choice reflects the observation that earthquake nucleation is more likely to occur at depth, consistent with previous studies suggest hypocentral depths of approximately 1 km for the Le Teil earthquake [10][13]. The input and data sample distribution are similar for y_{nuc} .



(a) Distribution of the along-strike nucleation position x_{nuc} . The comparison shows the prescribed normal distribution (mean = 0.5, standard deviation = 0.2) and the generated samples.

(b) Distribution of the along-dip nucleation position y_{nuc} . The comparison shows the prescribed uniform distribution between 0.5 and 1 and the generated samples.

Fig. 1.9 Input distributions for the nucleation position on the fault plane. The parameters x_{nuc} and y_{nuc} define the normalized along-strike and along-dip positions of the rupture initiation point, respectively. The figures compare the prescribed input distributions with the generated samples used in the stochastic rupture modeling.

Weak Zone Taper

1. Define the Weak Zone Taper

The weak zone taper is designed to increase the rise time and decrease the rupture velocity in the shallow fault region using a single parameter, the weak-zone-taper-strength $T \in [0, 1]$. Here, $T = 0$ corresponds to no tapering, while $T = 1$ applies the strongest tapering.

At the surface ($z = 0$), the scaling multipliers are defined as:

$$M_{\text{surface}, \tau_{\max}} = 1 + (\tau_{\max, \text{limit}} - 1)T, \quad (1.2)$$

$$M_{\text{surface}, V_{\text{rup}}} = 1 - (1 - V_{\text{rup}, \text{min}, \text{limit}})T, \quad (1.3)$$

where $\tau_{\max, \text{limit}}$ defines the maximum amplification of the rise time, and $V_{\text{rup}, \text{min}, \text{limit}}$ defines the maximum reduction of rupture velocity

The taper decreases linearly with depth, reaching unity at the taper depth z_{taper} , which defines the extent of the weak zone:

$$M(z) = \begin{cases} M_{\text{surface}} + (1 - M_{\text{surface}}) \frac{z}{z_{\text{taper}}}, & z \leq z_{\text{taper}}, \\ 1, & z > z_{\text{taper}}. \end{cases} \quad (1.4)$$

where z_{taper} defines the depth of the weak zone,

The taper is applied multiplicatively to the original rise time and rupture velocity fields:

$$\begin{aligned}\tau_{\max, \text{tapered}}(x, z) &= \tau_{\max, \text{original}}(x, z) M(z), \\ V_{\text{rup}, \text{tapered}}(x, z) &= V_{\text{rup}, \text{original}}(x, z) M(z).\end{aligned}\tag{1.5}$$

I assume a uniform distribution for the weak-zone taper strength T . Figure 1.10 shows the generated samples along with the prescribed input distribution. The taper depth is set to $z_{\text{taper}} = 0.5 \cdot W$, corresponding to the upper 50% of the fault width. The rise-time amplification factor is $\tau_{\max, \text{limit}} = 8$, which roughly doubles the rise time within the weak zone, while the rupture-velocity reduction factor is $V_{\text{rup}, \text{min}, \text{limit}} = 0.2$, reducing the rupture velocity on average to 80% of its original value in the weak zone.

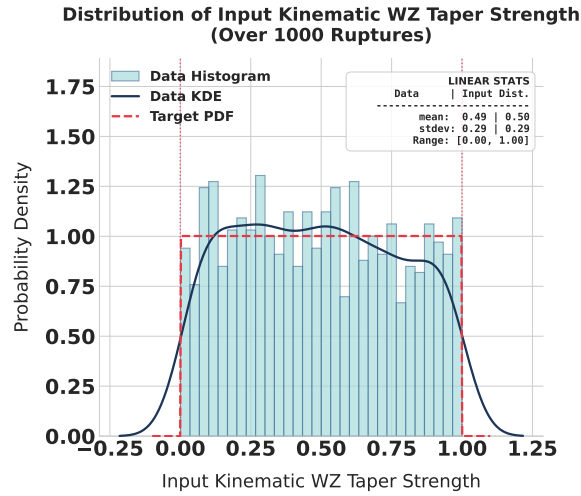


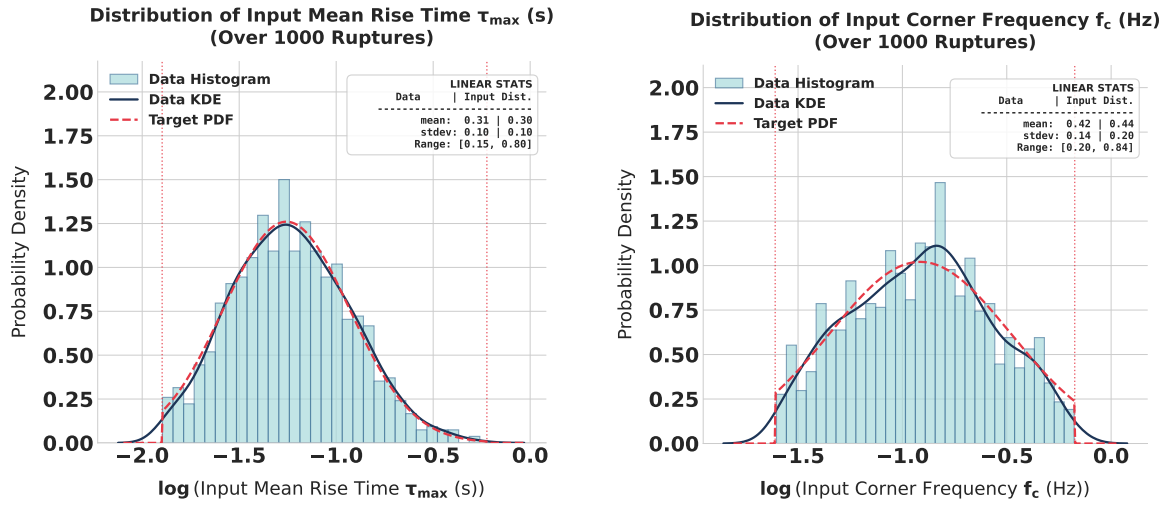
Fig. 1.10 Distribution of the weak-zone taper strength parameter T . The generated samples follow the prescribed uniform distribution between 0 and 1.

2. Impact of the Weak Zone Taper

Before examining the impact of the weak-zone taper, I first present the non-tapered distributions of the input mean rise time and corner frequency. Figure 1.11 shows the generated samples from 1000 stochastic rupture realizations alongside their prescribed truncated lognormal distributions. Specifically, panel (a) displays the mean rise time, while panel (b) shows the corner frequency, which is positively correlated with the mean rupture velocity.

A truncated lognormal distribution is adopted for the mean rise time, $\tau_{\max}$, with a mean of 0.3 s and a standard deviation of 0.1 s (linear scale). This yields a 68% confidence interval of 0.2–0.4 s [4]. The chosen mean value is based on the empirical scaling relationship between moment magnitude and rise time proposed by [17]. The distribution itself is bounded between 0.15 s and 0.8 s. Similarly, a truncated lognormal distribution is assumed for the corner frequency, f_c . The target mean is 0.44 Hz with a standard deviation of 0.2 Hz (linear scale),

corresponding to a 68% confidence interval of 0.26–0.66 Hz based on spectral analyses of seismological data [4].



(a) Generated samples and prescribed truncated lognormal distribution of the mean rise time $\tau_{\max}$.

(b) Generated samples and prescribed truncated lognormal distribution of the corner freq. f_c .

Fig. 1.11 Input distributions for the mean rise time and corner frequency used in stochastic rupture generation.

Figure 1.12 shows the tapered mean rise time, mean rupture velocity, and peak slip rate across the full fault depth. The peak slip rate is weakly correlated with rupture velocity.

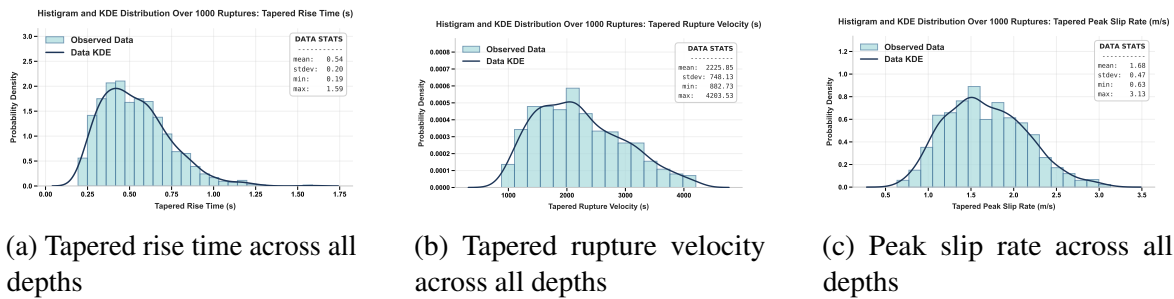


Fig. 1.12 Tapered rise time, rupture velocity, and peak slip rate for the entire fault depth. The peak slip rate shows a weak correlation with rupture velocity.

Figure 1.13 illustrates the separation of key source parameters—mean rise time, mean rupture velocity, and peak slip rate—between the weak zone and the surrounding hard rock. Within the weak zone, the mean rise time increases, whereas the rupture velocity and peak slip rate decrease relative to the hard rock. Since the total slip remains constant, the longer rise time naturally results in a lower peak slip rate. The lower peak slip rate is particularly important for lowering simulated PGA.

Table 1.1 summarizes the these source parameters affected by weak zone tapering, along with their statistical distributions for all depths, within the weak zone, and in the surrounding hard rock (outside the weak zone). The correlation coefficients between the slip field and the

mean rise time, mean rupture velocity, and peak slip rate follow the input values of 0.8 and 0.4, respectively, on the hard rock.

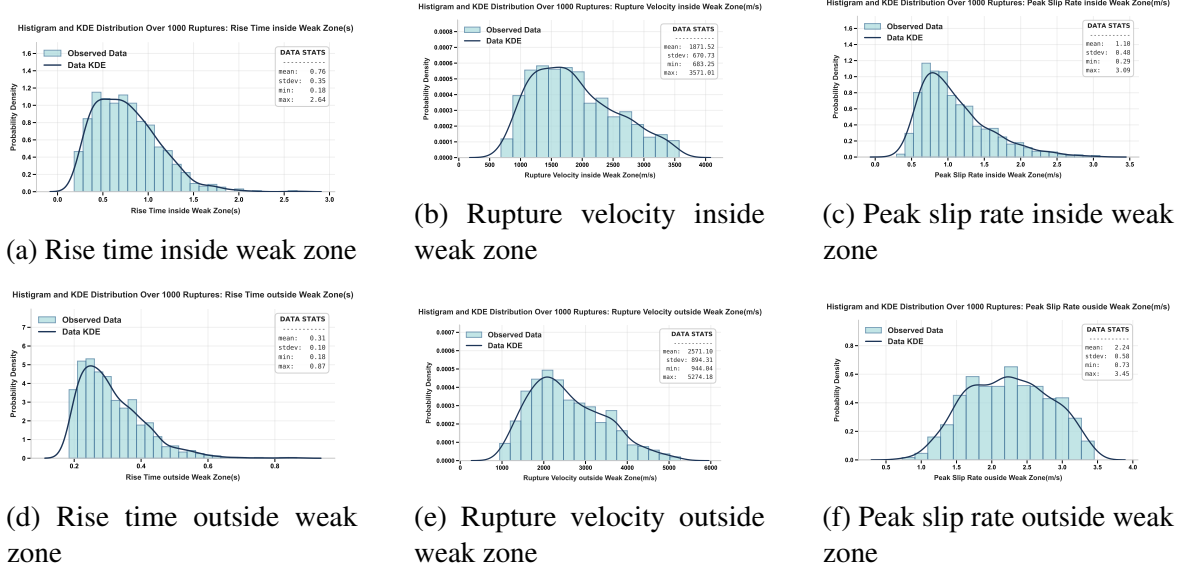


Fig. 1.13 Comparison of key rupture parameters inside (top row) and outside (bottom row) the weak zone. The weak zone exhibits increased rise time and decreased rupture velocity and peak slip rate relative to the surrounding hard rock.

Parameter	Mean Rise Time [s]	Mean Rupture Velocity [m/s]	PSR [m/s]	corr(slip, τ_{rise})	corr(V_{rup} , PSR)
All depths	0.54 ± 0.21	2206.23 ± 755.42	1.67 ± 0.48	0.25 ± 0.21	0.63 ± 0.12
Weak zone	0.78 ± 0.37	1852.73 ± 679.84	1.09 ± 0.48	0.30 ± 0.21	0.62 ± 0.18
Hard rock	0.31 ± 0.10	2550.67 ± 898.50	2.23 ± 0.58	0.77 ± 0.08	0.39 ± 0.16
Input (for hard rock)	0.30 ± 0.10	2460.54 ± 884.60	NA	0.80	0.40

Table 1.1 Comparison of key source parameters for all depths, the weak zone, and hard rock (outside the weak zone), as well as the input values. Values are reported as mean $\pm$ standard deviation. The input mean rupture velocity is calculated using Equation 1.1.

Moreover, the shape of the source time function differs between the weak zone and the surrounding hard rock. Within the weak zone, the source time function is more symmetric and bell-shaped.

Figure 1.14 shows an example from the first rupture scenario with a weak zone taper strength of $T = 0.61$. The source position is defined in coordinates along the fault width (W) and fault length (L). For example, $W = 128$ m and $L = 2089$ m indicates that the source is located 128 meters along the dip and 2089 meters along the strike. The fault corners are defined as follows: the upper left corner at (0, 0) m, the upper right corner at (0, 4200) m, the lower left corner at (1400, 0) m, and the lower right corner at (1400, 4200) m. As mentioned previously, the fault geometry has a length of $L = 4.2$ km and a width of $W = 1.4$ km. The parameter μ represents the mean rigidity across the fault plane, which is calculated from the input velocity model as the product of density and the square of the shear wave velocity.

The input velocity model is listed in Table 1.2. The rise time is the duration of the yoffe's source time function.

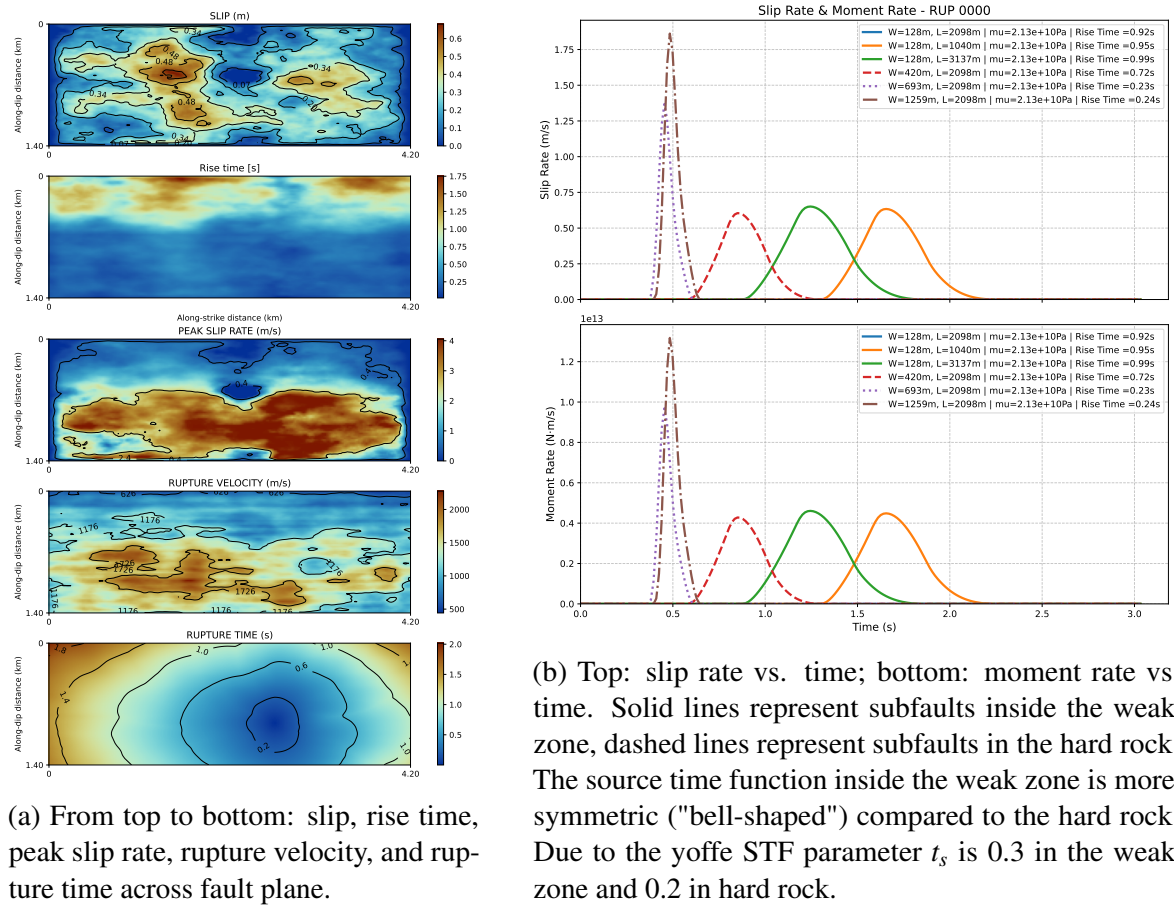


Fig. 1.14 Example of a rupture scenario with a weak zone taper strength of $T = 0.61$. Panel (a) shows each rupture parameter field on the fault plane, while panel (b) highlights the difference in source time function shapes between weak zone and hard rock.

1.2.3 Input Velocity model

For the simulations, two distinct velocity models are utilized: one representing hard rock conditions and the other representing weathered rock.

The baseline velocity model is derived from local passive and active surface wave measurements (inverted by Cécile Cornou). From the ten best-fitting inverted models, a mean model was calculated, with its parameters listed in Table 1.2. Figure 1.15 shows the individual inverted models in gray, the resulting mean model in black, and the velocity model used by [4] in red.

The newly inverted mean model is largely consistent with that of [4], except at shallow depths. This contrast in velocity around 20 meters, reflecting the transition between alluviums and marls, is important for accurately capturing soil resonance frequencies. These frequencies were measured from 184 H/V measurements (personal communication with Cécile Cornou).

Depth to top (m)	V_p (m/s)	V_s (m/s)	ρ (kg/m ³)	Q_p	Q_s
0.00	292.34	168.98	2000	29.23	16.90
4.75	612.04	353.78	2000	61.20	35.38
7.77	946.70	547.23	2000	94.67	54.72
17.58	3669.93	2121.35	2000	366.99	212.13
668.12	6278.89	3629.42	2000	627.89	362.94
1197.99	2090.10	1208.15	2000	209.01	120.82
1442.27	4065.27	2349.87	2000	406.53	234.99
2016.48	4120.47	2381.77	2000	412.05	238.18
2181.13	5939.80	3433.41	2000	593.98	343.34
5804.50	6169.14	3565.98	2000	616.91	356.60

Table 1.2 Baseline velocity model inverted by Cécile Cornou. The attenuation factors (Q_p and Q_s) are obtained by dividing V_p and V_s by 10.

As previously mentioned, the simulations employ two distinct model variations: a weathered rock model and a hard rock model, which are detailed in Table 1.3 and Table 1.4, respectively. Because density is not sensitive to the velocity model inversion process (personal communication with Cécile Cornou), it was initially fixed at a constant 2000 kg/m³ in the baseline model (Table 1.2). However, to better account for local geological units [4, 30, 13, 1] and rigidity, I adapted the depth-dependent density parameters for each layer based on [4].

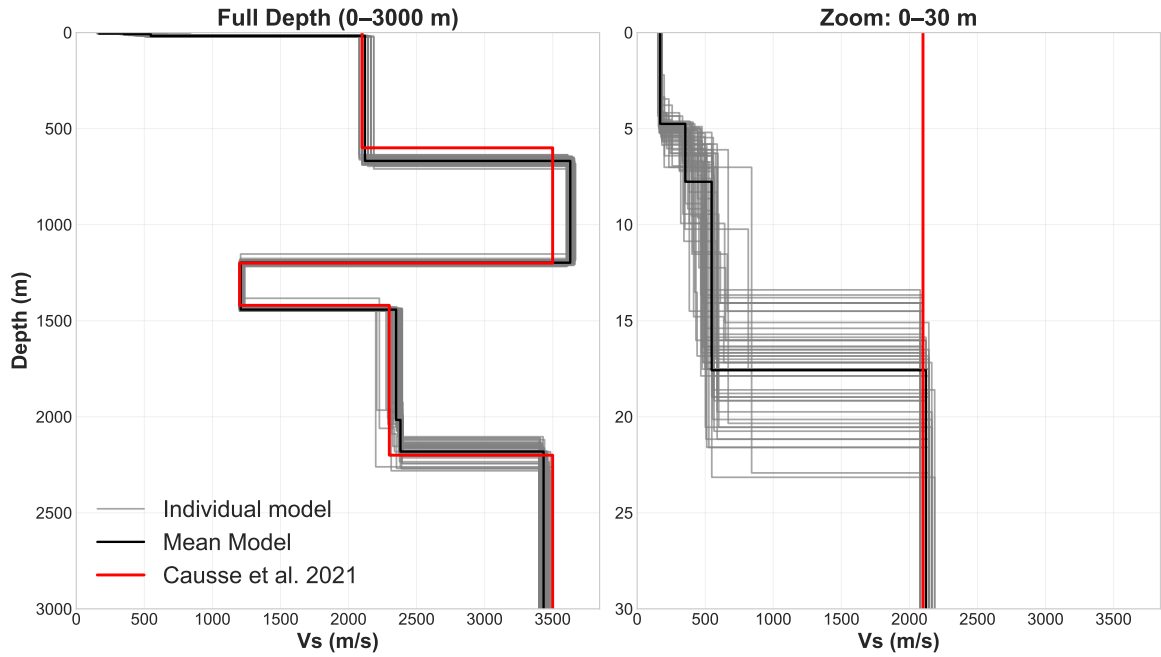


Fig. 1.15 Velocity models inverted using near-field ambient noise data (inversions by Cécile Cornou). The mean model is shown as a black line, and the ten best-fitting individual models are plotted in gray. The model used by [4] is plotted as a red line. The left panel shows the structure down to a depth of 3000 meters, covering the fault width, while the right panel provides a zoomed-in view of the near-surface structure (0–30 meters).

Layer	Depth to top (m)	V_p (m/s)	V_s (m/s)	ρ (kg/m ³)	Q_p	Q_s
1	0.00	1366.10	547.23	2000	136.61	54.72
2	17.58	3635.41	2121.35	2407	363.54	212.13
3	668.12	5995.80	3629.42	2688	599.58	362.94
4	1197.99	2024.68	1208.15	2165	202.47	120.82
5	1442.27	4015.95	2349.87	2465	401.59	234.99
6	2016.48	4627.07	2381.77	2470	462.71	238.18
7	2181.13	5900.69	3433.41	2470	590.07	343.34
8	5804.50	6075.39	3565.98	2470	607.54	356.60

Table 1.3 Weathered rock model. Attenuation factors Q_p and Q_s are defined as $V_p/10$ and $V_s/10$, respectively.

Impact of Input Velocity Model on Prior PGA at Graveyard

As mentioned in Section 1.2.3, two velocity models were used: a hard rock model and a weathered rock model. The weathered rock model includes a shallow layer ($V_s = 547\text{m/s}$) at approximately 20 meters depth to account for soil and structure resonance frequencies.

Table ?? shows that the horizontal median peak ground acceleration (PGA) increases by an absolute value of 3.51 m/s^2 , corresponding to a relative increase of approximately 51.92%, when using the weathered rock model compared to the hard rock model.

Layer	Depth to top (m)	V_p (m/s)	V_s (m/s)	ρ (kg/m ³)	Q_p	Q_s
1	0.00	3635.41	2121.35	2407	363.54	212.13
2	668.12	5995.80	3629.42	2688	599.58	362.94
3	1197.99	2024.68	1208.15	2165	202.47	120.82
4	1442.27	4015.95	2349.87	2465	401.59	234.99
5	2016.48	4627.07	2381.77	2470	462.71	238.18
6	2181.13	5900.69	3433.41	2470	590.07	343.34
7	5804.50	6075.39	3565.98	2470	607.54	356.60

Table 1.4 Hard rock model. I removed the first shallow layer from the weathered rock model to create this profile, making it similar to the reference model used by [4]. Attenuation factors Q_p and Q_s are defined as $V_p/10$ and $V_s/10$, respectively.

Model Type	Statistic	Max PGA Horizontal (m/s ²)	Max PGA Vertical (m/s ²)
Hard Rock (10 Hz)	Min	0.40	0.31
	Max	44.20	20.95
	Mean	8.63	3.99
	Median	6.76	3.25
	StDev	7.05	2.79
Weathered Rock (10 Hz)	Min	0.39	0.18
	Max	117.74	33.77
	Mean	13.61	3.63
	Median	10.27	2.91
	StDev	13.88	3.21

Table 1.5 Comparison of horizontal and vertical PGA parameters between Hard Rock and Weathered Rock velocity models (10 Hz cutoff).

Impact of Cutoff Frequency on Prior PGA at Graveyard

Seismograms were simulated up to 10 Hz, and the effects of different cutoff frequencies—5 Hz and 10 Hz—on the peak ground acceleration (PGA) are summarized in Table ???. Increasing the cutoff frequency from 5 Hz to 10 Hz results in a notable increase in both horizontal and vertical PGA values.

For the horizontal PGA, the median value increases from 5.26 m/s² at 5 Hz to 10.27 m/s² at 10 Hz, representing almost a twofold increase. Similarly, the vertical PGA shows an increase from 1.87 m/s² to 2.91 m/s². The standard deviation also grows substantially with higher cutoff frequency, indicating a wider variability in PGA values when higher frequency content is included in the simulations.

These results highlight the importance of selecting an appropriate cutoff frequency in ground motion simulations. Lower cutoff frequencies (e.g., 5 Hz) may underestimate the peak accelerations, especially for structures sensitive to high-frequency motion. Including higher frequency components (up to 10 Hz) captures more realistic ground motion amplitudes, which is particularly critical for evaluating structural response and seismic hazard in the study area.

Cutoff Frequency	Statistic	Max PGA Horizontal (m/s ²)	Max PGA Vertical (m/s ²)
5 Hz (weathered rock)	Min	0.31	0.17
	Max	38.01	15.33
	Mean	6.27	2.19
	Median	5.26	1.87
	StDev	5.00	1.59
10 Hz (weathered rock)	Min	0.39	0.18
	Max	117.74	33.77
	Mean	13.61	3.63
	Median	10.27	2.91
	StDev	13.88	3.21

Table 1.6 Comparison of summary statistics for the simulated maximum horizontal and vertical Peak Ground Acceleration (PGA). Results are presented for both 5 Hz and 10 Hz cutoff frequencies using a weathered rock velocity profile.

Impact of Site Location (on hanging wall or footwall) on Prior PGA at Graveyard

The influence of site location relative to the fault plane was examined by comparing the graveyard site positioned on the hanging wall (HW) versus the footwall (FW). Table ?? summarizes the PGA statistics for both scenarios using the 10 Hz soil model.

The results indicate that the hanging wall location generally experiences slightly higher horizontal PGA values than the footwall, with a median horizontal PGA of 10.27 m/s² for HW compared to 9.79 m/s² for FW. The vertical PGA is somewhat similar between the two locations, although the footwall shows a slightly higher median vertical PGA (3.28 m/s²) than the hanging wall (2.91 m/s²). Maximum PGA values are slightly higher on the hanging wall, reflecting the typical “hanging wall effect” observed in near-fault ground motions.

Overall, these results highlight that site location relative to the fault plane can influence the amplitude of ground motion, particularly in the horizontal direction. Accounting for HW and FW effects is therefore important for accurate seismic hazard assessment and structural design in near-fault regions.

Site Location (Soil 10 Hz)	Statistic	Max PGA Horizontal (m/s ²)	Max PGA Vertical (m/s ²)
Hanging Wall (HW)	Min	0.39	0.18
	Max	117.74	33.77
	Mean	13.61	3.63
	Median	10.27	2.91
	StDev	13.88	3.21
Footwall (FW)	Min	0.00	0.00
	Max	112.18	27.06
	Mean	12.75	3.97
	Median	9.79	3.28
	StDev	12.77	3.25

Table 1.7 Comparison of PGA parameters between Hanging Wall (HW) and Footwall (FW) locations (10 Hz Soil Model).

Impact of Site Elevation on Prior PGA at Graveyard

Finally, the impact of site elevation on simulated ground motion was assessed. This analysis is particularly relevant because the simulations include maximum frequencies ($f_{\max}$) up to 10 Hz, and the minimum shear-wave velocity ($V_{s,\min}$) in the input velocity model is 547 m/s. The subfault size was set to 27 meters, based on the criterion:

$$\text{subfault dimension} = \frac{V_{s,\min}}{2f_{\max}}$$

This means that the shallowest subfaults are located very close to the site. Even after applying a 5% slip taper at the fault edges, the proximity of the source could potentially influence the simulated PGA. To evaluate this effect, site elevations of 0 meters and 30 meters were tested.

Table 1.8 summarizes the resulting PGA statistics for both elevations using the 10 Hz soil model. The results show that increasing the station elevation from 0 m to 30 m reduces the simulated horizontal PGA amplitudes. The median horizontal PGA decreases from 10.27 m/s² at 0 m elevation to 7.00 m/s² at 30 m elevation, while the mean PGA decreases from 13.61 m/s² to 8.30 m/s². The maximum horizontal PGA is also significantly reduced, from 117.74 m/s² to 68.92 m/s².

The vertical PGA values also show noticeable reductions with increasing elevation. The median vertical PGA increases slightly from 2.91 m/s² to 4.56 m/s², while the mean vertical PGA increases from 3.63 m/s² to 5.37 m/s². However, the maximum vertical PGA remains comparable, decreasing only slightly from 33.77 m/s² to 32.63 m/s². Overall, these results indicate that station elevation influences near-field simulated PGA, likely because increasing the elevation changes the effective source-to-site distance for the shallowest subfaults.

Station Elevation (Weathered Rock 10 Hz)	Statistic	Max PGA Horizontal (m/s ²)	Max PGA Vertical (m/s ²)
0 Meters	Min	0.39	0.18
	Max	117.74	33.77
	Mean	13.61	3.63
	Median	10.27	2.91
	StDev	13.88	3.21
30 Meters	Min	0.42	0.40
	Max	68.92	32.63
	Mean	8.30	5.37
	Median	7.00	4.56
	StDev	6.01	3.64

Table 1.8 Comparison of PGA parameters between 0 m and 30 m station elevations for the 10 Hz soil model.

Balancing Model Affordability and Functionality

Based on the sensitivity tests conducted in this section, it is evident that the cutoff frequency (yielding a median peak ground acceleration [PGA] of 5.26 m/s² at 5 Hz compared to 10.27 m/s² at 10 Hz) and the inclusion of a minimum shear wave velocity layer ($V_s = 547$ m/s) at a depth of approximately 20 meters (yielding a median PGA of 6.76 m/s² on hard rock compared to 10.27 m/s² on weathered rock) are the most significant contributors to the amplitude of the simulated ground motions. Furthermore, additional tests demonstrated that the location of the site relative to the fault also influences the results, a site situated on the footwall produced a median PGA of 9.79 m/s², compared to 10.27 m/s² on the hanging wall and the two tested locations are 60 meters apart. As for the elevation of the site (topographic effect), the median horizontal PGA decreases from 10.27 m/s² at 0 m elevation to 7.00 m/s² at 30 m elevation, while the mean PGA decreases from 13.61 m/s² to 8.30 m/s².

However, in my opinion, the impact of these highly localized site parameters cannot be rigorously concluded without considering a highly detailed micro-structural model of the shallow subsurface. For example, Figure 1.16 illustrates the shallow structure of a strand of the San Andreas Fault [29]. This figure demonstrates how micro-structural details can impact the velocity structure within a 20-meter depth. Achieving this level of precision requires the integration of multiple disciplines, including geodesy (3D topographic modeling), seismology (3D velocity tomography, attenuation of seismic waves at different frequencies, and seismic anisotropy), and high-resolution geological modeling informed by laboratory experiments and fracture mechanics. However, for the study area in Le Teil, France, the high-resolution data required to build a model with this level of micro-structural detail is currently unavailable, even if it is available, it would be computationally expensive. Such extreme detail is arguably unnecessary. When considering the balance between model functionality and affordability—specifically regarding computational cost and data acquisition—an analogy to commercial navigation systems (e.g., Google Maps) is highly relevant. It would be

computationally absurd and highly inefficient for a navigation app to render every granular physical detail of a street simply to route a user from home to the office. Similarly, in my earthquake models, it is not strictly necessary to capture every micro-scale geological variation in the shallow soil structure. By focusing on the most influential parameters, I can bypass the need for unattainable data resolution while still producing a predictive, statistically sound model. Therefore, I have standardized the forward modeling in the subsequent sections to exclusively use a 10 Hz cutoff frequency combined with the weathered rock velocity profile. Given the limitations of the existing regional datasets at the time of writing, I argue that the selected input parameters represent the most robust and optimized configuration possible.

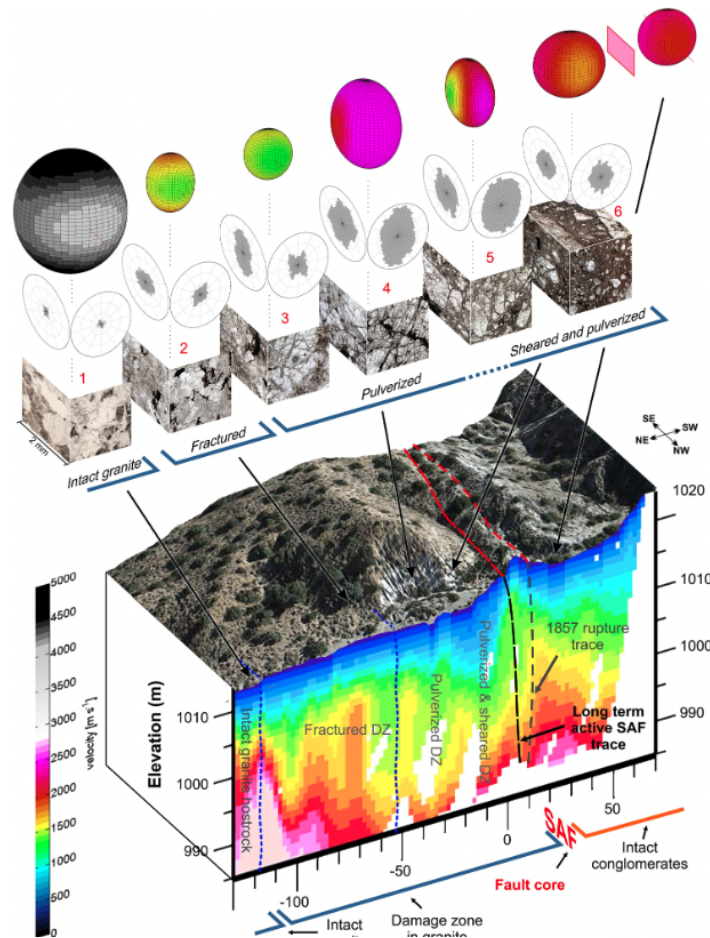


Fig. 1.16 Summary of results from laboratory measurements, microstructure analysis, and in situ seismic measurements, demonstrating the implications for different damage-type zones. Ellipses, whose sizes scale with the absolute velocities (see color scheme), illustrate velocity anisotropy. They were obtained by interpolating between the three known directions. Different damage-type zones inferred from laboratory velocities and microstructures transfer well to the in situ velocities (Source: [29]).

1.3 Overturned Rock Site

The overturning of the rock implies that the localized vertical ground acceleration exceeded $1g$ [4]. Near-fault strong motion patterns have also been qualitatively investigated using displaced rocks for other seismic events; a similar phenomenon was observed following the 2019 Ridgecrest, California earthquakes, where modeling the displacement of broken rocks indicated a PGA exceeding $1g$ [23]. In another study, Michael et al., (2002) [24] estimated the minimum horizontal acceleration required to initiate observed rock displacements for 1999 Hector Mine, California, earthquake.

To apply this physical constraint to the simulations, the vertical Peak Ground Acceleration (PGA) was calculated at station 0 (the location of the overturned rock). For each rupture, the PGA values were first averaged across eight repeated model runs. The dataset was then filtered, retaining only those ruptures where the averaged downward vertical PGA exceeded the $1g$ threshold. Figure 1.17 presents the acceleration time series for Rupture 0078 as a representative example. Notably, the upward acceleration peak does not need to exceed $1g$; for the rock to lose contact with the surface and overturn, the ground must simply accelerate downward faster than gravity.

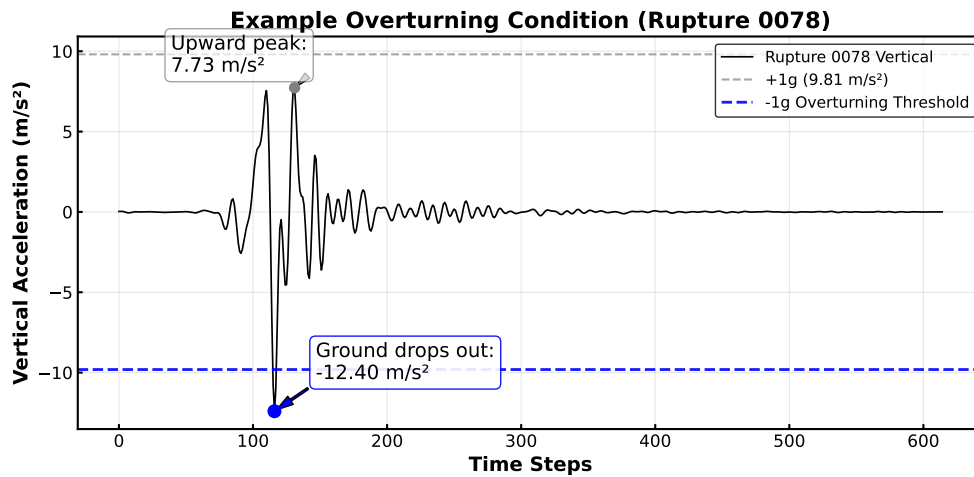


Fig. 1.17 Acceleration time series for Rupture 0078, demonstrating a downward vertical PGA that exceeds the $1g$ threshold required to overturn the rock.

Figure 1.18 compares the prior (unfiltered) and posterior (filtered) PGA distributions at station 0. Applying this threshold resulted in a retention rate of approximately 40%, meaning that I rejected 60% of the simulated ruptures based solely on this criterion.

Figure 1.19 illustrates the impact of this filtering process on the five input source parameters. I observed two primary changes: first, the distribution of the mean rise time shifts toward lower values; second, the corner frequency (f_c) shifts toward higher values, which corresponds to a higher mean rupture velocity. The distributions of the remaining input parameters do not exhibit any significant shift.

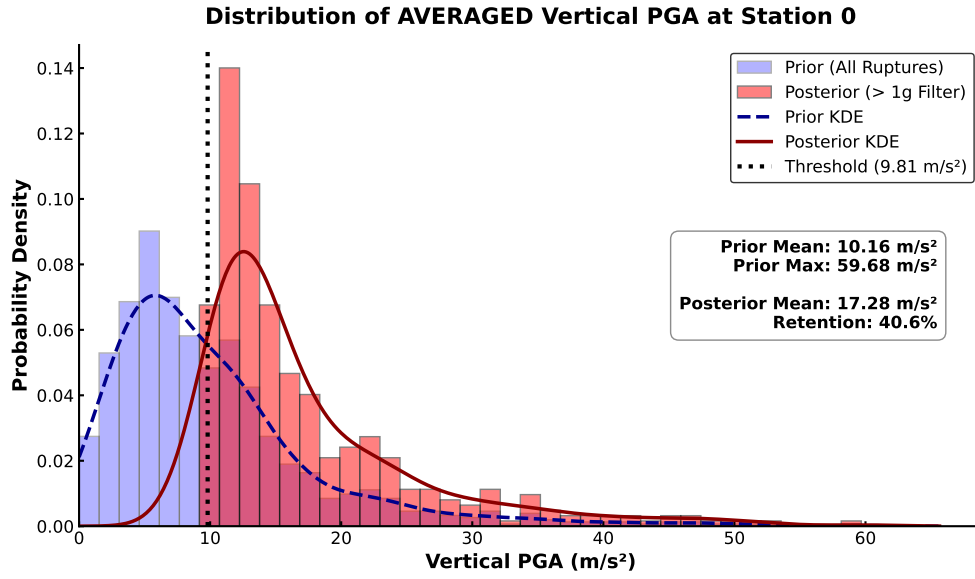


Fig. 1.18 Comparison of the prior and posterior vertical PGA distributions at station 0.

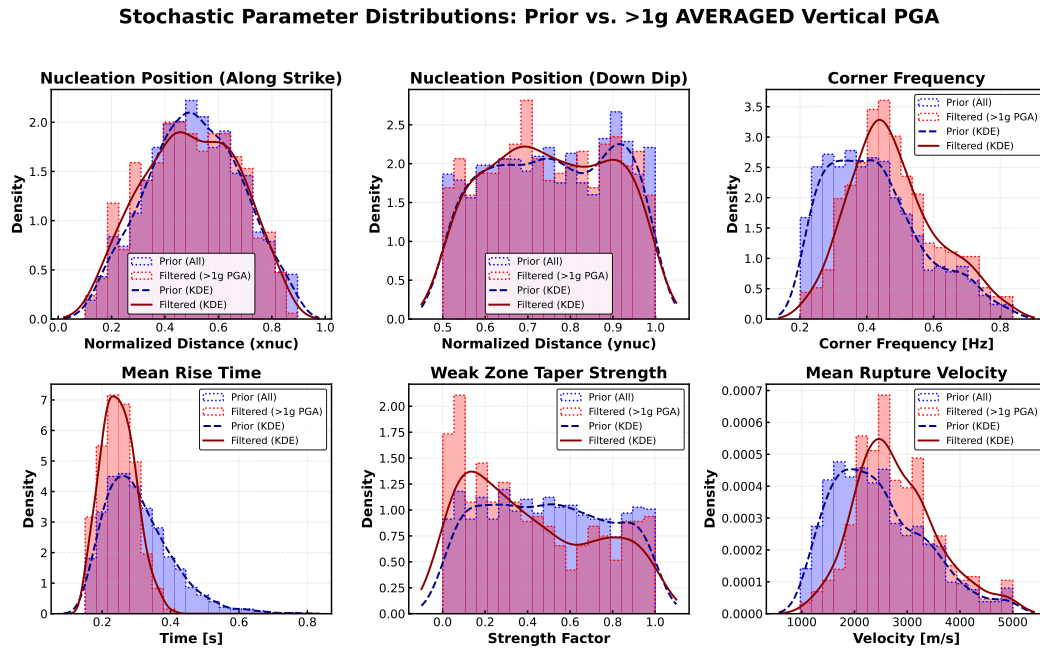


Fig. 1.19 Prior versus posterior distributions of the macroscopic kinematic source parameters. The near-identical distributions indicate that the overturned rock observation alone does not provide sufficient constraint to meaningfully update the source model.

1.4 Damaged Buildings

Following the earthquake, building damage was observed in the nearby city of Le Teil [31, 6]. Initial observations were categorized by firefighters using a green-yellow-red tagging system based on damage severity. These tags were later converted to damage grades according to the EMS-98 scale [16]. The definitions of the damage grades according to the EMS-98 scale are presented in Table 1.9. Expert judgment was applied to determine the probability that a building assigned a specific color tag corresponds to a particular damage grade. The correspondence between damage tags and EMS-98 damage grades for this event is detailed by [35].

Damage Grade	Severity	Description of Physical Damage
Damage Grade 1	Negligible	Hairline cracks in walls; small pieces of plaster fall.
Damage Grade 2	Moderate	Small cracks in walls; fall of fairly large pieces of plaster.
Damage Grade 3	Substantial	Large and extensive cracks in most walls; roof tiles slip.
Damage Grade 4	Very Heavy	Serious failure of walls; partial structural failure.
Damage Grade 5	Destruction	Total or near-total collapse of the building.

Table 1.9 Definition of Damage Grades according to the EMS-98 Scale [16].

In this thesis, I rely exclusively on the damaged building data from the AFPS emergency survey, which corresponds to the DD1 dataset detailed by [35]. Figure 1.20 from [35] illustrates the percentage of firefighter color tags that were converted to each EMS-98 damage grade. Table 1.10 presents the proportion of total buildings assigned to each damage grade, comparing the direct conversions derived from field observations against those refined by expert opinion. In the case of a total inventory of 306 buildings, the estimated count of buildings in DG1 based purely on field observations is approximately 73 (0.24×306). In contrast, applying the expert opinion proportions adjusts this count to approximately 110 (0.36×306).

Building tag	Damage grade	Count	Percentage (%)
Green	1	91	61
Green	2	22	15
Green	3	35	24
Yellow	3	95	90
Yellow	4	8	8
Yellow	5	2	2
Red	4	47	64
Red	5	27	36

Fig. 1.20 Conversion percentages from rapid assessment color tags to EMS-98 damage grades (Source: [35]).

Damage Grade	Field Observations	Expert Opinion
1	0.24	0.36
2	0.06	0.22
3	0.42	0.32
4	0.18	0.09
5	0.09	0.01

Table 1.10 Comparison of building proportions assigned to each damage grade based on field observations versus expert opinion.

Next, I performed forward modeling using an existing artificial neural network (ANN) model [20] to predict the number of buildings in each damage grade. Within this ANN framework, the mean building damage is estimated based on the mechanical model proposed by [21]. The ANN requires three primary input parameters: the ratio of the building's resonance frequency to the soil's resonance frequency, the ground motion intensity measures on hard rock (PGA/PGV), and the H/V amplitude (A_{0HV}), which serves as a proxy for soil amplification. Figure 1.21 illustrates the conceptual layer structure of the ANN, detailing its specific inputs and outputs. For the PGA/PGV input, I utilized the source parameters introduced in the previous section, applying a cutoff frequency of 10 Hz on weathered rock. The modeling results are presented in Figures 1.22 and 1.23. The analysis for the input parameters for the ANN is detailed in 1.4.1

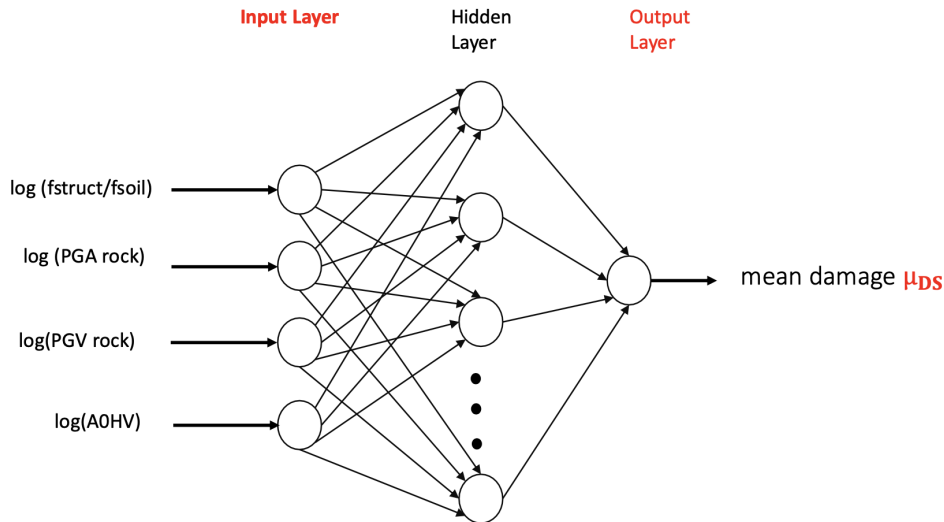


Fig. 1.21 Conceptual layer structure of the ANN model, detailing the specific input parameters and the resulting outputs (Source: Cécile Cornou).

Figure 1.22(a) illustrates the number of buildings assigned to each damage grade. Black stars denote the field observations derived directly from the conversion of firefighter tags, while orange diamonds represent expert opinions. The figure displays the predicted mean for each damage grade (DG) along with its $\pm 2\sigma$ uncertainty bounds. Notably, the predicted

mean underestimates DG2 building counts while overestimating DG3. This result aligns with the findings of [35] and is likely driven by dataset inconsistencies arising when rapid damage assessments (red–yellow–green tags) are converted into EMS-98 damage grades [11, 26]. Therefore, I combined DG2 and DG3 and present the updated results in Figure 1.23(a).

Figure 1.22(b) displays the correlation matrix for the predicted damage grades DG1–DG5, revealing a high degree of correlation among DG2, DG3, DG4, and DG5. Similarly, Figure 1.23(b) shows the correlation matrix for the predicted damage grades DG1, DG2&3, DG4, and DG5. In both cases, DG1 is negatively correlated with the other damage grades. This indicates that a lower number of buildings classified as DG1 corresponds to a higher number of buildings assigned to more severe damage states.

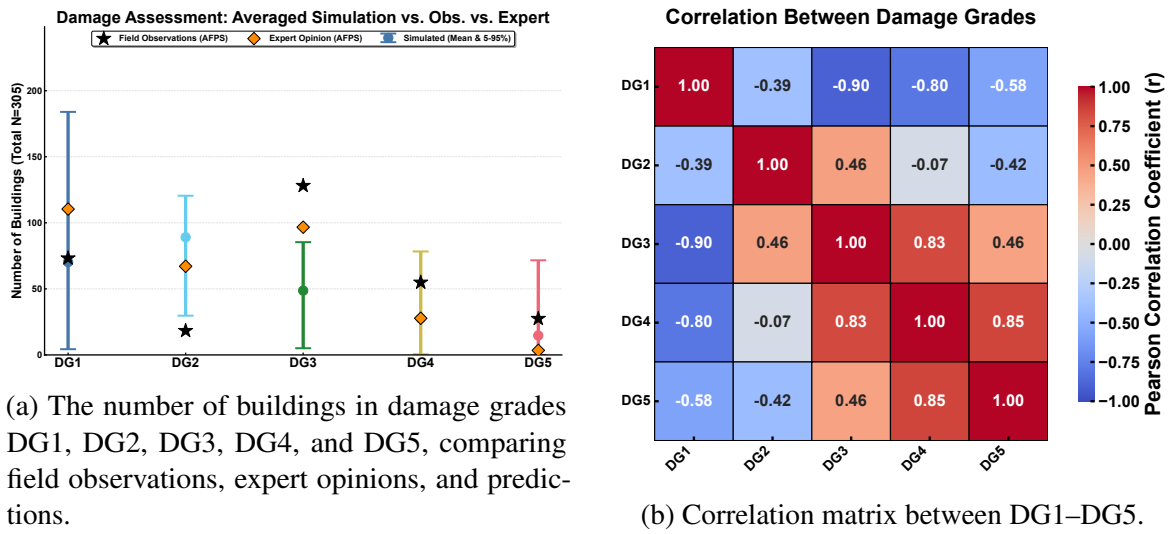


Fig. 1.22 Comparison of observed and predicted building damage. (a) Damage summary comparing field observations (firefighter tags), expert opinions, and model predictions across 1,000 simulated ruptures. (b) Correlation matrix showing the relationships and dependencies between the predicted damage grades (DG1–DG5).

Figure 1.24, adapted from [11], highlights the application of expert judgment following the 2023 Kahramanmaraş, Turkey earthquakes. The pie chart displays the percentage of originally assigned damage grades that were either downgraded or promoted, emphasizing the subjectivity and uncertainty inherent in this assessment process. These results justify the combination of DG2 and DG3. The purpose of this combination is not merely to fit the observations better, but rather to resolve the inherent ambiguity within the assessment data itself.

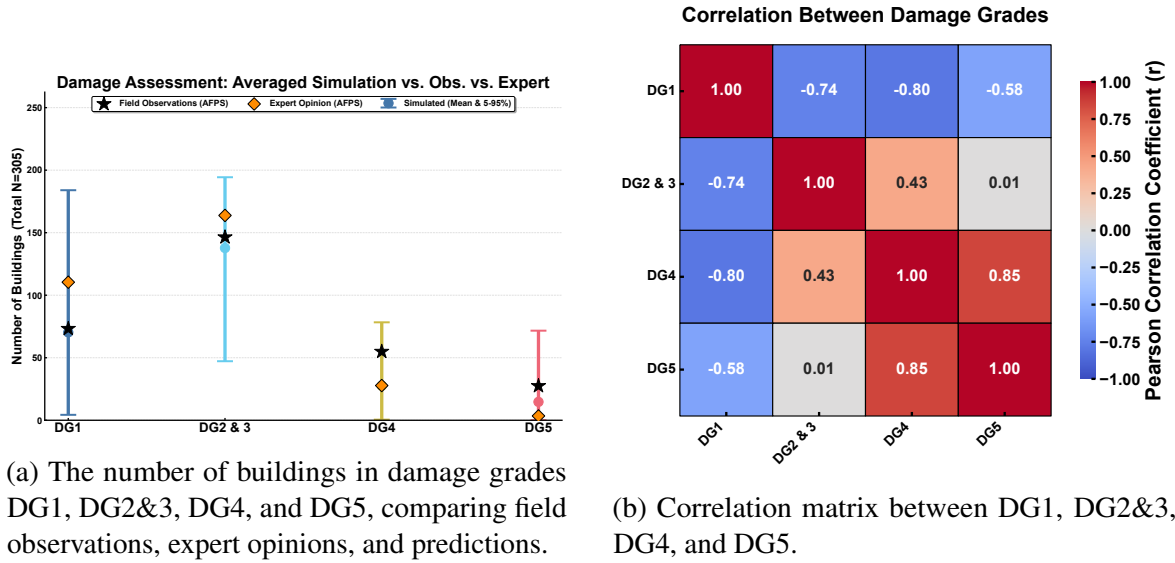


Fig. 1.23 Comparison of observed and predicted building damage. (a) Damage summary comparing field observations (firefighter tags), expert opinions, and model predictions across 1,000 simulated ruptures. (b) Correlation matrix showing the relationships and dependencies between the predicted damage grades (DG1, DG2&3, DG4, DG5).

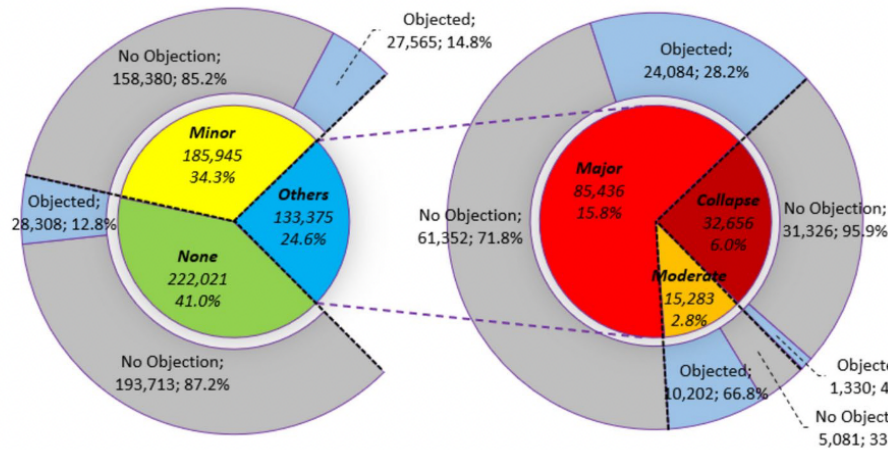


Fig. 1.24 Reassessment of building damage grades following the Turkey earthquake, demonstrating the proportion of downgraded and promoted grades (Source: [11]).

1.4.1 Input Parameters for Damage Prediction with ANN

Following the earthquake, 485 on-site building inspections were conducted by the French Association of Earthquake Engineering (AFPS) [33]. Using the French National Building Database (BNDB), 306 of these observed buildings were successfully associated with a specific construction date and height. To determine the total building height, the number of floors for each structure was multiplied by an assumed story height of 2.8 meters. Furthermore, a structural classification was assigned to each building based on a comprehensive French structural dataset, following the methodology proposed by [25].

Figure 1.25 presents the spatial distribution of the metadata utilized as inputs for the ANN. The soil resonance frequencies (f_0) were derived from 184 active and passive H/V measurements provided by Cécile Cornou (Figure 1.25a). The building resonance frequency (f_{bat}), shown in Figure 1.25(b), is calculated as a function of the building's height, which in turn is derived from the number of floors (Figure 1.25c). The amplitude of the H/V spectral ratio (A_{0HV}), also measured by C. Cornou, is utilized as a proxy for site amplification (Figure 1.25d). The building-to-soil resonance frequency ratio ($R_{bs} = f_{struct}/f_0$) is mapped in Figure 1.25(e). Finally, the building topology is classified into two primary categories: masonry and reinforced concrete (Figure 1.25f).

Figure 1.26 displays the mean predicted damage, over 1,000 simulations, plotted against the median input PGA, median input PGV, site amplification factor (A_{0HV}), and frequency ratio (R_{bs}). It is important to note that for any given site, A_{0HV} and R_{bs} remain constant across all 1,000 ruptures (each simulation is averaged by 8 repeats to remove random phases).

The 306 evaluated buildings are categorized into two primary structural typologies: masonry and reinforced concrete. This structural classification heavily dictates the model's damage predictions. For instance, all reinforced concrete buildings exhibited a predicted mean damage of zero across all 1,000 simulations.

When analyzing the entire dataset in aggregate, the input PGA, PGV, and R_{bs} show no overarching linear correlation with predicted damage. However, when isolating specific building typologies, strong relationships emerge. Most notably, for masonry buildings, the input A_{0HV} is highly correlated with the predicted mean damage, yielding a Pearson's correlation coefficient of $r = 0.91$.

Finally, to investigate the influence of input ground motion parameters on individual structures, Figure 1.27 illustrates the predicted damage versus the input PGA for two specific buildings across the 1,000 realizations. One building is centrally located within the building inventory (denoted as 'Center', Figure 1.27a), while the other is situated near the graveyard (Figure 1.27b). The predicted damage for both buildings exhibits a strong positive correlation with the input prior PGA, yielding Pearson's correlation coefficients of $r = 0.90$ and $r = 0.82$, respectively.

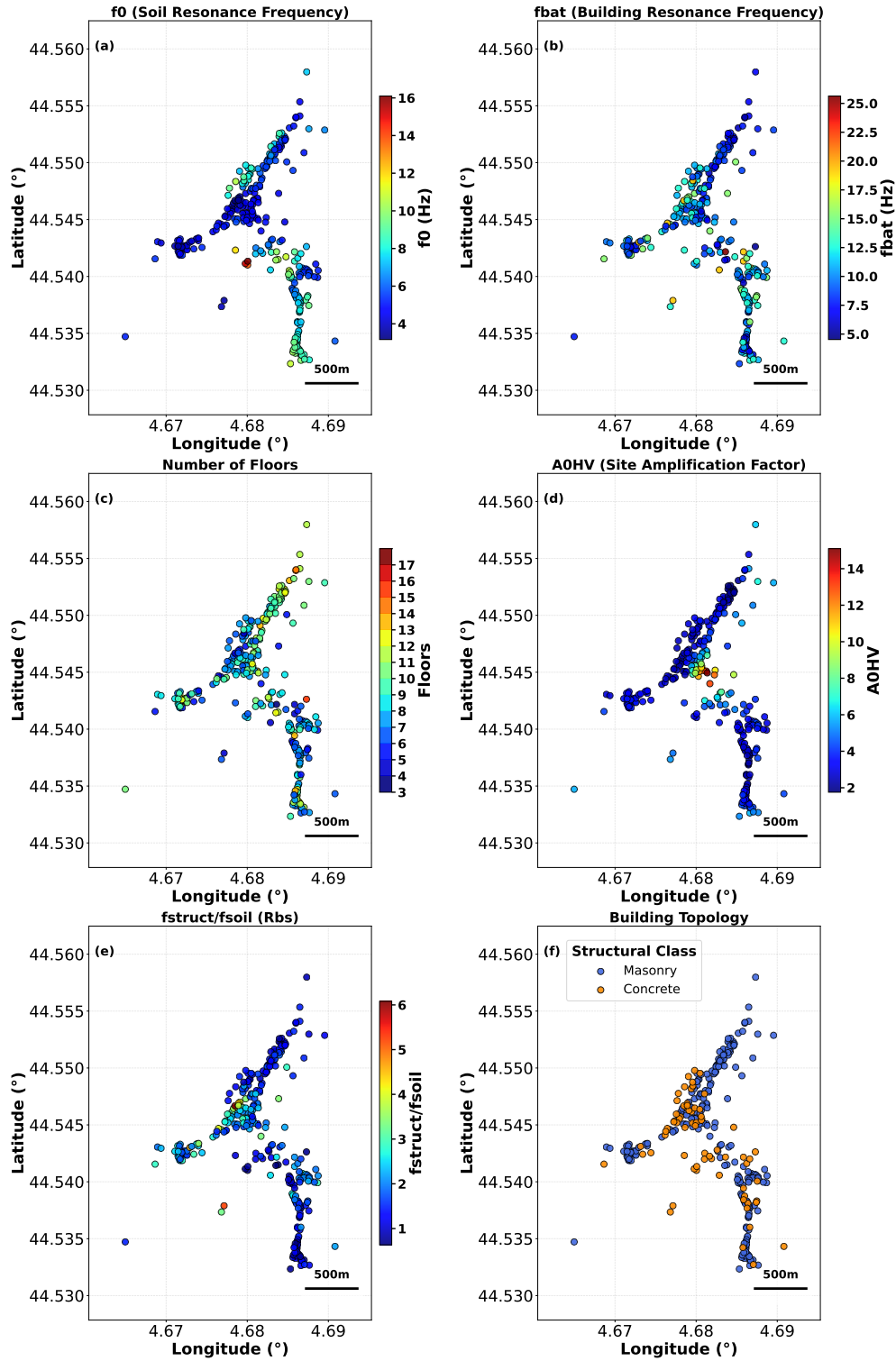


Fig. 1.25 Spatial distribution of building and site characteristics in the study area, displayed across six panels: (a) soil fundamental frequency (f_0), (b) building resonance frequency (f_{bat}), (c) number of building floors, (d) site amplification factor (A_{0HV}), (e) building-to-soil resonance frequency ratio ($R_{bs} = f_{bat}/f_0$), and (f) building topology indicating the structural class (Masonry vs. Concrete).

Correlation of Input Parameters vs. Predicted Mean Damage

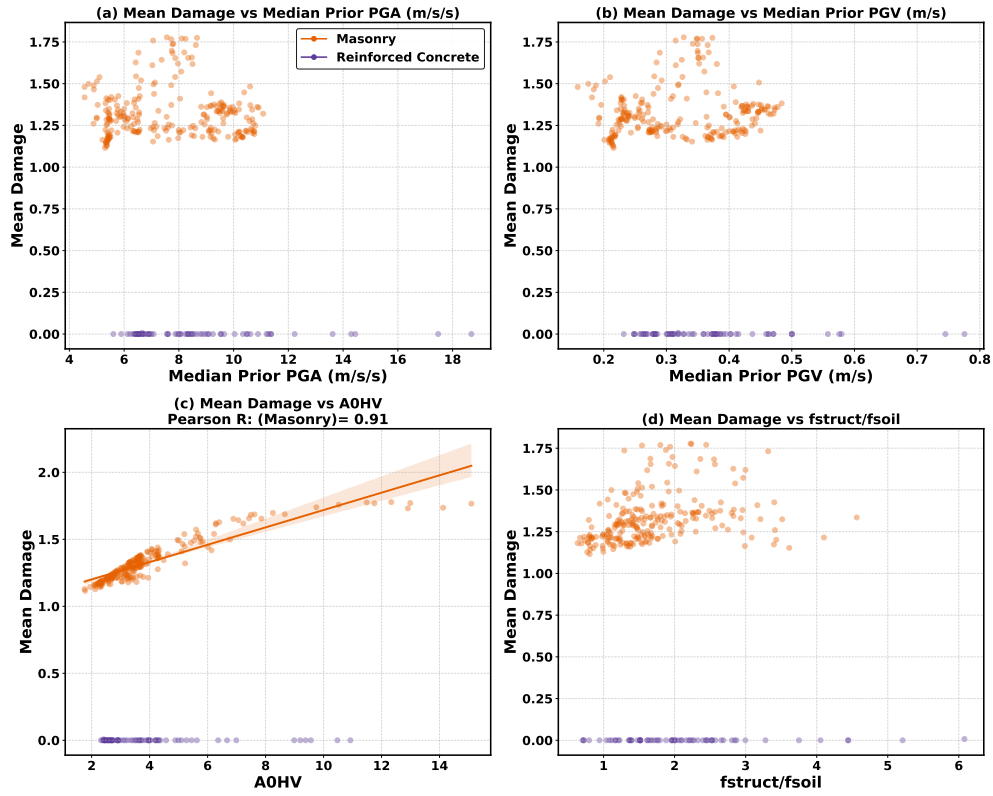


Fig. 1.26 Correlation plots showing the mean predicted damage over 1,000 simulations against median input PGA, PGV, A_{0HV} , and R_{bs} , categorized by structural topology (masonry vs. reinforced concrete).

Prior PGA vs Damage over 1000 Realizations at Graveyard and Center

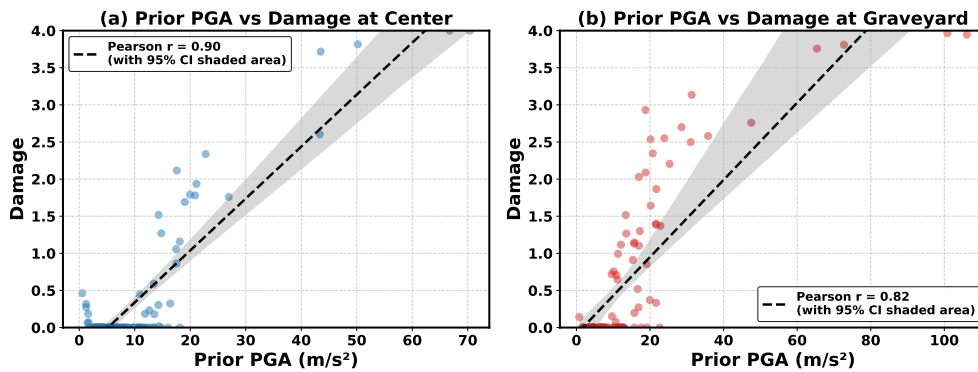


Fig. 1.27 Predicted damage versus input prior PGA across averaged 1,000 realizations for two selected buildings: (a) a centrally located building ('Center') and (b) a building located near the graveyard.

1.4.2 Methodology for Bayesian Update

A critical component of this chapter is updating the simulated ground motion scenarios (e.g., Peak Ground Acceleration) using real-world, non-instrumented observations. A fundamental

challenge when using building damage data is that these macroscopic indicators are physically correlated. For instance, a simulated rupture that over-predicts severe structural collapse (DG5) will systematically under-predict slight damage (DG1). Treating these observations as independent datasets in a standard Bayesian framework would result in severe statistical bias and artificially over-constrain the posterior distribution.

To address this limitation, I implement a Multivariate Normal (MVN) likelihood function to evaluate the scenarios. I propose a covariance approach that enforces custom, trust-weighted standard deviations to strictly define the acceptable error margins.

To evaluate the simulations fairly, I construct a hybrid covariance matrix (Σ) that respects the physical interdependencies of the damage grades while enforcing targeted strictness based on the reliability of the field observations. Let k denote the number of active macroscopic damage grades utilized in the updating process.

First, I calculate the empirical covariance matrix ($\mathbf{S}$) directly from the N simulated rupture outputs. To isolate the physical relationship between the damage variables, I strip away the empirical variances to obtain the pure correlation matrix ($\mathbf{R}$). For any two damage grades i and j , the empirical correlation is defined as:

$$R_{ij} = \frac{S_{ij}}{\sigma_{\text{emp},i} \sigma_{\text{emp},j}} \quad (1.6)$$

where $\sigma_{\text{emp},i}$ represents the empirical standard deviation of damage grade i across the entire simulation catalog. To understand the importance of the correlation matrix $\mathbf{R}$, consider the physical trade-offs between damage states. If a simulated earthquake is stronger than the actual event, it will naturally predict more collapsed buildings (DG5) and fewer lightly damaged buildings (DG1), because those buildings shifted into higher damage states. Because of this structural reality, simulation errors are not independent; rather, they form a correlated, elliptical distribution. The matrix $\mathbf{R}$ captures this precise relationship, mathematically defining the “tilt” of the acceptable error bounds. By incorporating $\mathbf{R}$, the likelihood function evaluates errors contextually: a simulation is penalized less if its inaccuracies align with logical physics (e.g., over-predicting severe damage while proportionally under-predicting slight damage). Conversely, a simulation is penalized heavily if its errors contradict structural mechanics (e.g., over-predicting both slight and severe damage simultaneously).

Rather than arbitrarily scaling the final likelihood scores to represent my confidence in certain datasets, I adjust the strictness of the Bayesian update by directly scaling the standard deviations. Because the damaged building targets in this study rely on expert opinion—which provides a deterministic building count rather than a statistical distribution with a known variance—I define a base acceptable standard deviation, $\sigma_{\text{base},i}$. This defaults to 10 buildings, or 5 buildings if the target count is very small (e.g., for DG5, the expert opinion gives a target of 3.4 buildings, so I use a standard deviation of 5 instead of 10).

When field observations are less reliable, I introduce a trust weight, $w_i \in (0, 1]$, for each damage grade. For example, the combined damage grade dataset (DG2&3) carries a higher

degree of uncertainty for two reasons. First, intermediate damage grades are inherently susceptible to visual misclassification—a building initially tagged as DG2 might actually belong in DG1, while a DG3 structure might be severely damaged enough to promote to a DG4 classification. Second, combining these grades yields a larger total building count. Applying the strict base standard deviation ($\sigma_{\text{base}} = 10$) to this much larger combined pool would over-constrain the model. Therefore, to account for both observational uncertainty and the larger target size, the custom standard deviation is calculated as:

$$\sigma_{\text{custom},i} = \frac{\sigma_{\text{base},i}}{\sqrt{w_i}} \quad (1.7)$$

For instance, applying a lower trust weight of $w_i = 0.3$ to the combined DG2&3 dataset adjusts the acceptable error margin, yielding a custom standard deviation of $10/\sqrt{0.3} \approx 18.3$.

This inverse-square-root relationship ensures that high-confidence datasets (e.g., $w_i = 1.0$) maintain a narrow, strict probability density envelope. Conversely, assigning a lower trust weight expands the standard deviation, effectively widening the target zone and preventing highly uncertain field data from disproportionately rejecting plausible rupture scenarios.

Next, I define $\mathbf{D}_{\text{strict}}$ as a $k \times k$ diagonal matrix where the diagonal entries are $\sigma_{\text{custom},i}$. The final hybrid covariance matrix, Σ , is assembled via matrix multiplication:

$$\Sigma = \mathbf{D}_{\text{strict}} \mathbf{R} \mathbf{D}_{\text{strict}} \quad (1.8)$$

For any specific element within the matrix, this expands to $\Sigma_{ij} = R_{ij} \sigma_{\text{custom},i} \sigma_{\text{custom},j}$. This resulting matrix governs the MVN likelihood function, coupling my tightly controlled error margins with the structural correlation of the ground motion simulations.

With the hybrid covariance matrix established, I evaluate the plausibility of each simulated rupture. Let $\mathbf{x}$ be a $k \times 1$ vector representing the simulated building damage counts for a specific rupture scenario, and let $\boldsymbol{\mu}$ be a $k \times 1$ vector containing the observed field targets. The likelihood $L(\mathbf{x})$ of observing the simulated damage given the target field data is evaluated using the Multivariate Normal probability density function:

$$L(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right) \quad (1.9)$$

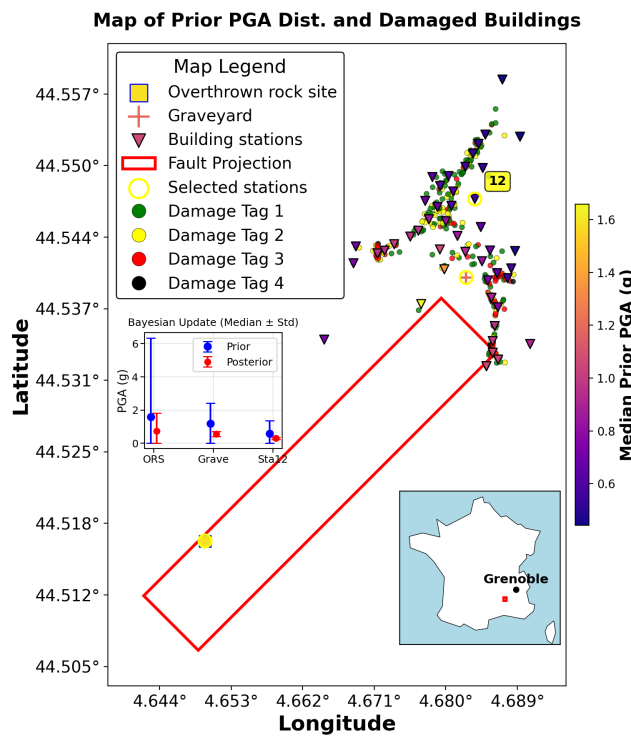
Within this formulation, the exponent term represents the squared Mahalanobis distance. By utilizing the inverse of the hybrid covariance matrix (Σ^{-1}), this distance metric effectively standardizes the simulation errors and accounts for covariance, penalizing orthogonal deviations much more strictly than collinear deviations.

The final step in this updating phase is to normalize the likelihood scores into relative weights that can be applied to the predictive variables (e.g., localized PGA at specific stations). For a total catalog of N simulations, the posterior weight W_j for simulation j is calculated as:

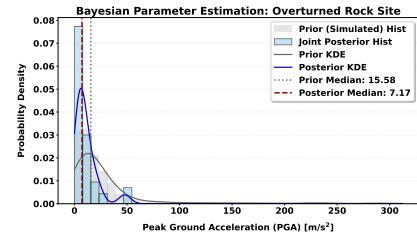
$$W_j = \frac{L(\mathbf{x}_j)}{\sum_{m=1}^N L(\mathbf{x}_m)} \quad (1.10)$$

These normalized weights ($\sum_j W_j = 1.0$) dictate the posterior probability mass of each ground motion scenario. Scenarios that align well with the correlated damage observations—bounded by the trust-weighted standard deviations—dominate the posterior distribution, while mathematically improbable scenarios are effectively filtered out of the final hazard assessment.

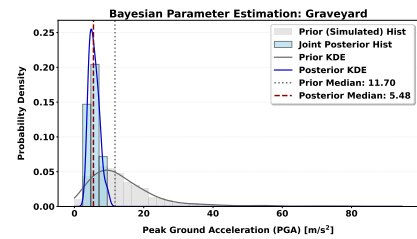
Figure 1.28 presents a map detailing the building damage investigations, displaying the severity tags assigned by firefighters alongside virtual stations colored by their median prior PGA. The figure highlights three specific sites of interest (circled in yellow) and their corresponding Bayesian update results. Specifically, Figure 1.28(a) shows the overturned rock site, which shifts from a prior median PGA of 15.58 m/s^2 to a posterior median of 7.17 m/s^2 . Figure 1.28(b) illustrates the graveyard site, shifting from a prior median of 11.70 m/s^2 to a posterior median of 5.48 m/s^2 . Finally, Figure 1.28(c) displays a more distant station, which shifts from a prior median of 5.62 m/s^2 to a posterior median of 2.88 m/s^2 .



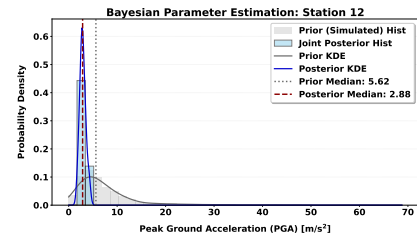
Map of damaged buildings colored by firefighter observation tags [33], virtual stations colored by median prior PGA across 1,000 ruptures, and the fault projection. Three key stations are circled in yellow.



(a) Prior and posterior median PGA for the overturned rock site.



(b) Prior and posterior median PGA for the graveyard site.



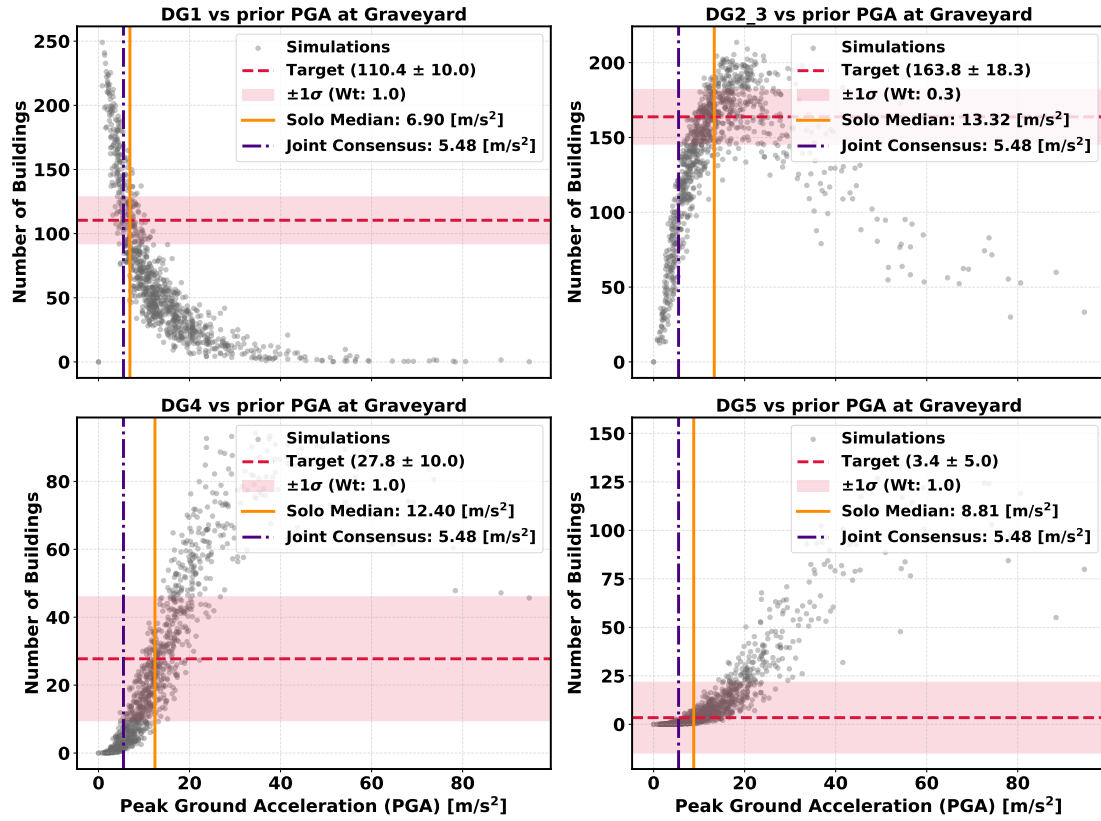
(c) Prior and posterior median PGA for station 12.

Fig. 1.28 Prior median PGA map (left) and corresponding Bayesian posterior distributions for three selected stations using building damage observations (right).

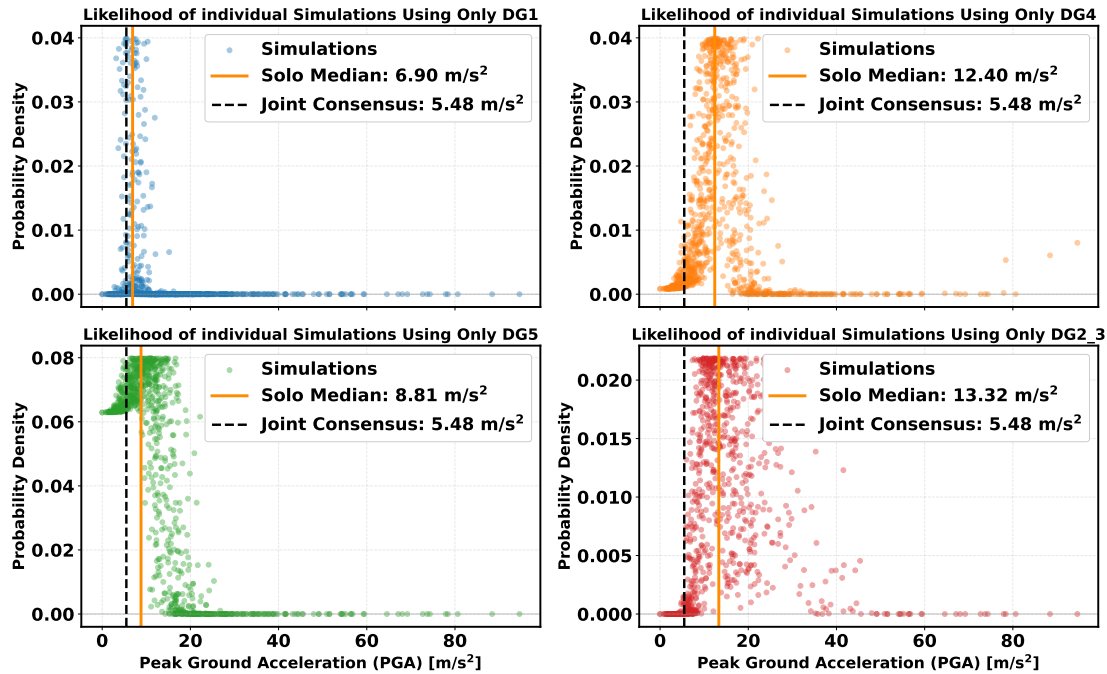
For example, I show the diagnostic plot for the graveyard to illustrate better my bayesian update strategy. Figure 1.29 illustrates the predicted number of buildings versus the prior PGA across approximately 1,000 averaged realizations at the graveyard site. The standard deviation is strictly set to 10 buildings (or customized per the logic above) because the target is a deterministic value derived from expert opinion. For context, the targets are: 110.4 buildings for DG1, 27.8 buildings for DG4, 163.8 buildings for the combined DG2&3 dataset, and 3.4 buildings for DG5. The lower panel of the figure displays the corresponding likelihood weights calculated using Equation 1.10. The red shaded zone indicates the area spanned by the weighted standard deviation, representing the parameter space that yields the highest statistical weight.

Figure 1.30 visualizes the individual likelihood influences and the joint MVN consensus for the graveyard site. In this scenario, the four individual influences are the observed building counts in the DG1, DG4, combined DG2&3, and DG5 categories. The Maximum Likelihood Estimate (MLE)—representing the single rupture with the highest statistical weight—places the PGA at 6.67 m/s^2 for the graveyard site. Conversely, the median consensus derived from the fully updated posterior distribution is estimated at 5.48 m/s^2 .

When combining these individual observations, the joint likelihood remains high only where all datasets overlap. To demonstrate this filtering effect across 1,000 ruptures: 194 achieved a normalized relative likelihood $> 1\%$ for DG1 ($w = 1.0$), 829 for DG4 ($w = 1.0$), 761 for DG5 ($w = 1.0$), and 756 for DG2&3 ($w = 0.3$). However, when constrained by the joint MVN likelihood across all four categories, only 48 out of the 1,000 ruptures retained a relative likelihood greater than 1%.



(a) Fragility diagnostic results, showing the predicted building counts for DG1, DG2&3, DG4, and DG5 as a function of prior PGA at the graveyard site.



(b) Likelihood weights for each individual damage grade.

Fig. 1.29 Comparison of fragility diagnostics and raw weight distributions derived from the Bayesian update.

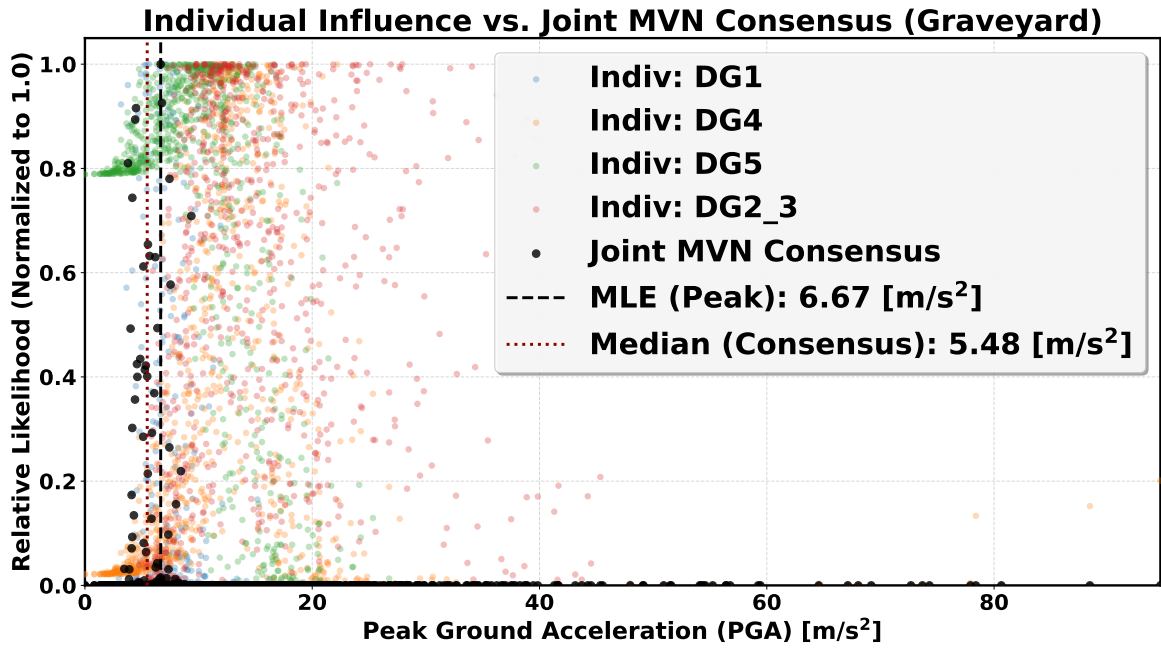


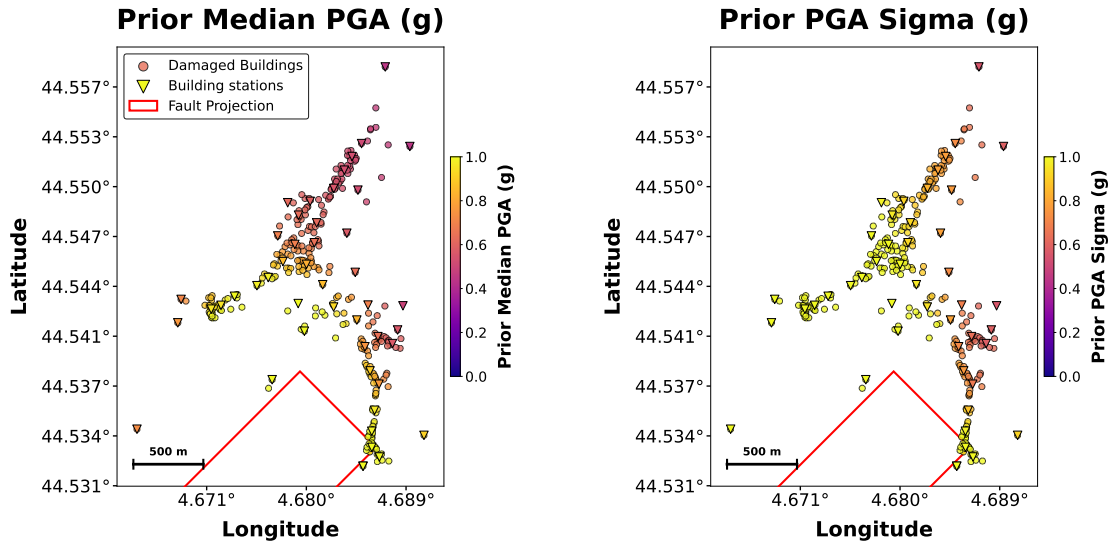
Fig. 1.30 Diagnostics of the Multivariate Normal (MVN) likelihood update at the graveyard site (Station 1), highlighting the individual influences of the damage grade observations alongside the joint consensus distribution.

1.4.3 Bayesian Update on Prior PGA for all Stations

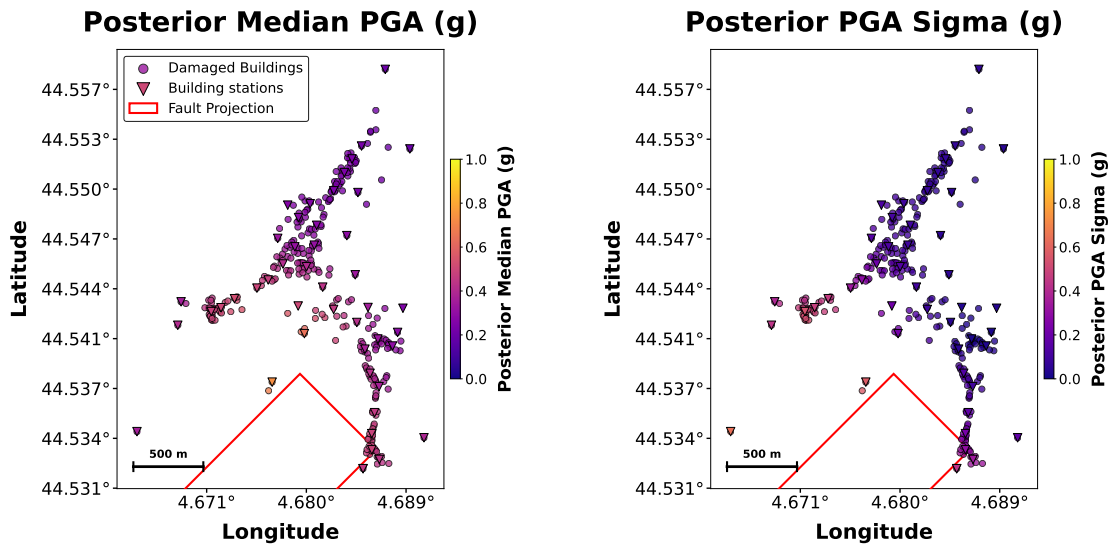
Figure 1.31 presents a spatial comparison of the prior and posterior PGA distributions, displaying the median values alongside their standard deviations. The prior model exhibits a median PGA of 0.95 ± 0.37 g across the study area. Following the Bayesian update, the posterior median PGA is significantly reduced to 0.13 ± 0.17 g. This substantial reduction indicates that the unconstrained prior simulations systematically overpredicted the ground motion amplitudes. To further quantify these reductions, Figure 1.32 provides a spatial difference map comparing the posterior models directly to the priors, revealing a mean reduction of $52.03\% \pm 3.04\%$. This result is based on 1,000 ruptures, with each rupture repeated 8 times, yielding a total of 8,000 simulations.

To investigate whether the input source parameters could be constrained by the building damage observations, I performed a Random Forest sensitivity analysis, as shown in Figure 1.33. Because both the test R^2 and Out-of-Bag (OOB) R^2 scores are negative, the algorithm failed to identify a predictive relationship. This indicates that the kinematic source parameters cannot be meaningfully constrained using this dataset alone.

Nevertheless, figure 1.34 illustrates the prior and posterior marginal distributions for the five input source parameters and the rupture velocity following the Bayesian update. Consistent with the results of the Random Forest tests, no parameters exhibit significant shifts from their respective priors.



(a) Prior spatial distribution: Median (calculated spatially over 49 stations) of the prior median PGA (across 8,000 ruptures) is 0.95 g with a standard deviation of 0.37 g.



(b) Posterior spatial distribution: Median PGA of 0.13 g with a standard deviation of 0.17 g.

Fig. 1.31 Comparison of the prior (a) and posterior (b) median PGA and standard deviation maps.

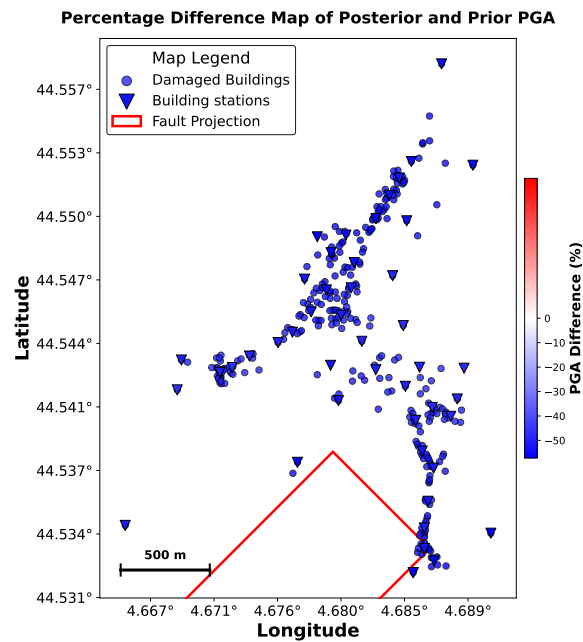


Fig. 1.32 PGA difference map showing a mean reduction of $52.03\% \pm 3.04\%$ of the median PGA at each station.

Random Forest Parameter Sensitivity on Averaged Measurables

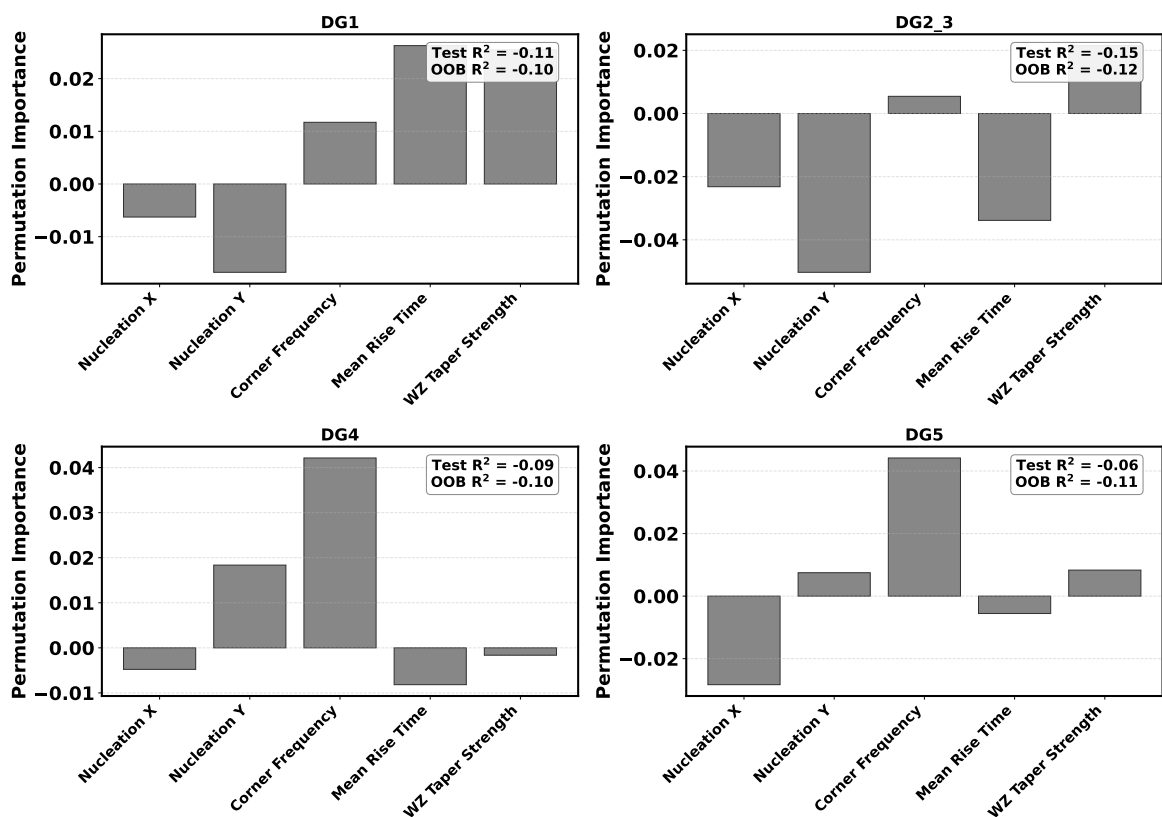


Fig. 1.33 Random Forest sensitivity analysis for DG1, DG2&3, DG4, and DG5 against the five input source parameters. Both the test R^2 and OOB R^2 scores are negative, indicating no predictive relationship.

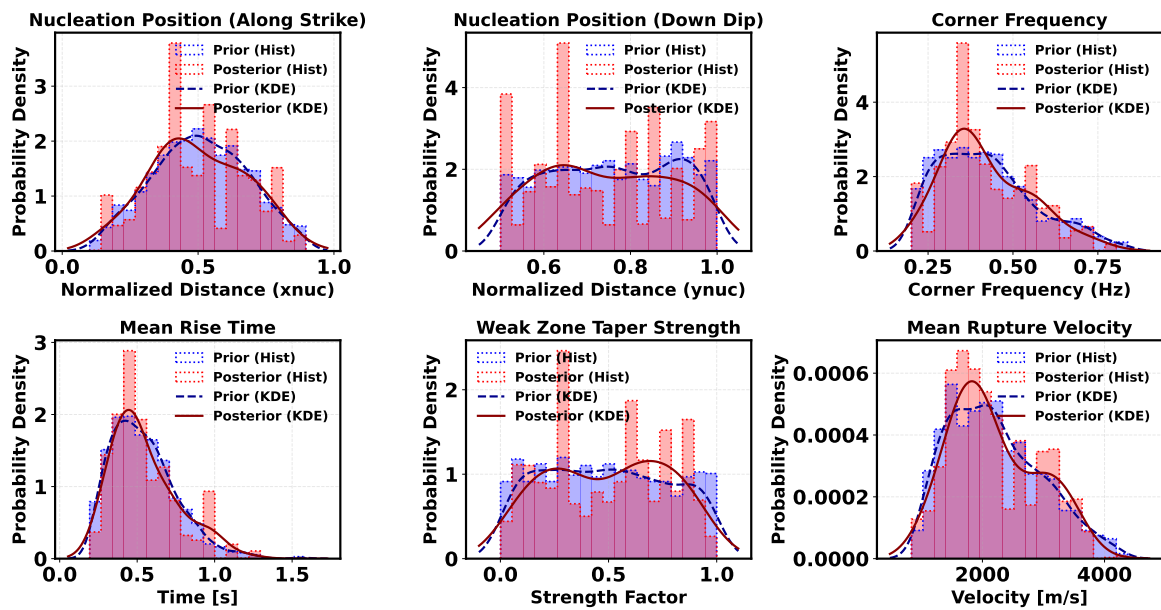


Fig. 1.34 Comparison of prior and posterior marginal distributions for the five input kinematic source parameters and rupture velocity following the Bayesian update using building damage observations. No significant posterior updating is observed.

1.5 Grave Slab Modelling

To use the graveyard observations, I model slab movement using a Coulomb friction model (Causse et al., 2021) and compare the simulated motion with field observations. This analysis evaluates the relative displacement of slabs with respect to the ground / walls, the azimuth of the final slab positions, and the polarity of displacement in opposite directions. The modeling relies on several key assumptions: (1) slabs are loosely placed on the ground; (2) the fault location relative to the graveyard is accurate; (3) slabs slide without toppling; and (4) the retaining wall / headstone does not move relative to the ground.

Figure 1.35 presents a comparison between a compiled observational dataset of slab displacements toward the NW and SE directions (left panel) and the corresponding numerical predictions (right panel). The simulated results represent an average of 8,000 total simulations, consisting of 1,000 rupture realizations. The NW and SE cases are modeled in pairs; specifically, for every imposed seismogram, I simultaneously modeled both the NW and SE slab movements by alternating the retaining wall position. To account for contact uncertainty, the friction coefficient (μ) for each simulation was randomly sampled from discrete values of 0.1, 0.2, 0.3, or 0.4. These discrete values are grounded in laboratory experiments on slab movement conducted by M. Causse.

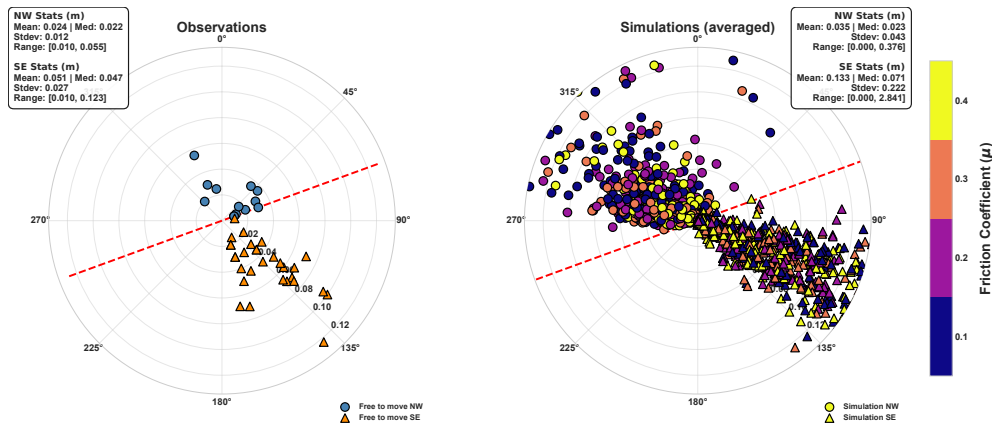


Fig. 1.35 Comparison of observed and simulated slab displacements. The simulated results are averaged 1000 ruptures (8,000 simulations in total), with the friction coefficient (μ) for each run randomly sampled from 0.1, 0.2, 0.3, and 0.4.

Figures 1.36 and 1.37 present the detailed statistical distributions of the 1000 rupture scenarios. Subfigures (a) and (b) display the input horizontal PGA in logarithmic and linear scales, respectively, which is naturally identical for both movement directions. Subfigures (c) and (d) compare the simulated slab displacements and azimuths against the real-world observations. Figure 1.36 focuses on slabs moving toward the NW, while Figure 1.37 focuses on slabs moving toward the SE.

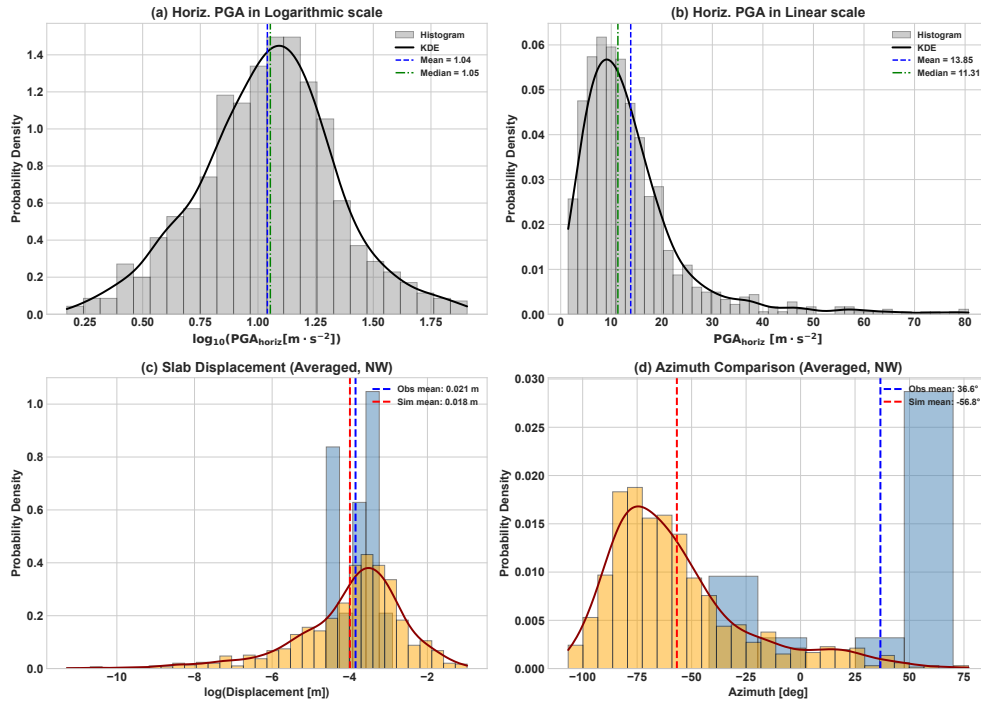


Fig. 1.36 Distributions of input ground motion and resulting slab kinematics toward the NW direction across 1000 ruptures. Panels (a) and (b) show the input horizontal PGA on logarithmic and linear scales. Panels (c) and (d) compare the simulated relative slab displacements and movement azimuths against field observations for the NW direction.

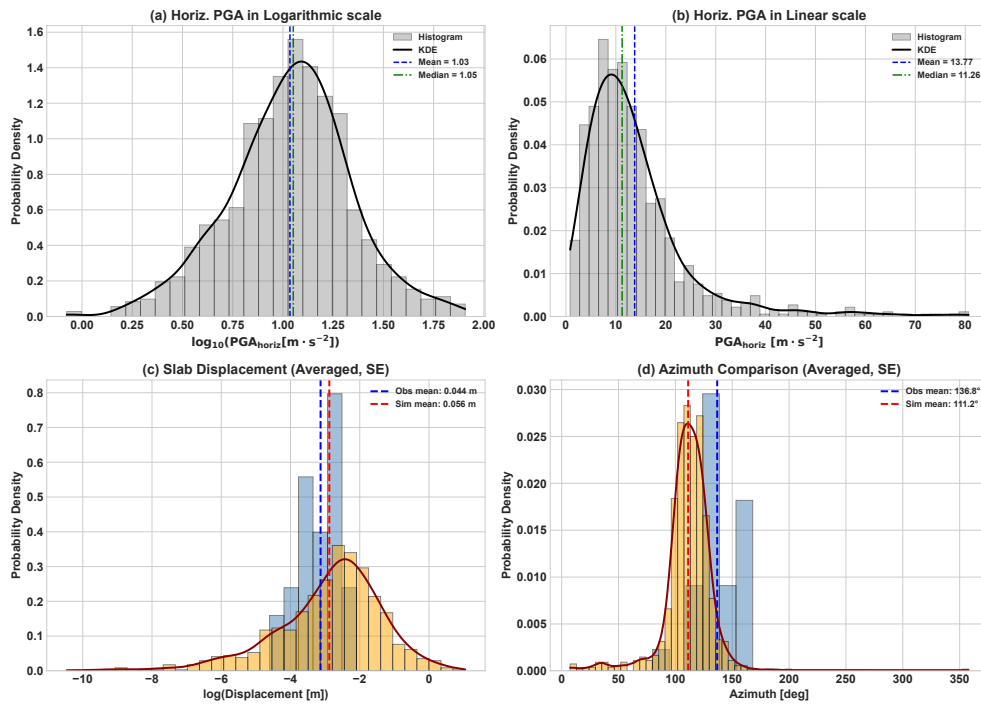


Fig. 1.37 Distributions of input ground motion and resulting slab kinematics toward the SE direction across 1000 ruptures. Panels (a) and (b) show the identical input horizontal PGA on logarithmic and linear scales. Panels (c) and (d) compare the simulated relative slab displacements and movement azimuths against field observations for the SE direction.

1.5.1 Forward Modeling of Slab Displacement

To simulate the dynamic response and residual displacement of rigid objects subjected to seismic ground shaking, I used a Coulomb friction modelling proposed by [4]. the model evaluates the absolute and relative kinematics of a rigid block resting on a frictional surface, driven by three-component ground motion. It accounts for dynamic variations in normal force, lift-off (detachment) mechanics, and inelastic boundary collisions (e.g., adjacent walls).

Input and Output Parameters

The numerical model requires several physical and kinematic inputs, including a three-component seismogram to describe the ground movement as a function of time with a sufficient integration time step. Another key input is the mass of the slab (M), which is calculated assuming dimensions of $2 \text{ m} \times 1.2 \text{ m} \times 0.2 \text{ m}$ and a material density of 2500 kg/m^3 . The static friction coefficient (μ) between the slab and the ground is randomly sampled from a discrete set of values (0.1, 0.2, 0.3, and 0.4) for each rupture scenario. Additionally, the model incorporates boundary conditions defined by the azimuth angle (θ) and the spatial orientation (e.g., Southeast or Northwest) of a stationary retaining wall (headstone), which acts as an inelastic barrier. Based on these inputs, the model generates two primary outputs: the final residual displacement magnitude and vector (EW and NS coordinates) after the ground motion ceases, and a time-series kinematic status flag (ISTATUS) that tracks the physical state of the block at each time step (1: Stuck, 2: Sliding, 3: Detached/Gapping, 4: Wall Contact).

Vertical Dynamics and Normal Force (Lagrange Method)

Because vertical ground accelerations significantly alter the frictional capacity of the block, the model dynamically computes the effective normal force (F_N) at each time step. Using the Lagrange multiplier method, the block is assumed to remain in contact with the ground unless the vertical inertial forces exceed gravity.

The vertical ground acceleration $\ddot{u}_Z(t)$ is derived from the numerical differentiation of the input vertical velocity. The time-dependent normal force is calculated as:

$$F_N(t) = M(g + \ddot{u}_Z(t)) \quad (1.11)$$

where g is the acceleration due to gravity.

During intense vertical shaking, the ground may accelerate downward faster than gravity (i.e., $\ddot{u}_Z(t) < -g$). When this occurs, $F_N(t)$ becomes negative, indicating that the block has lost contact with the ground. To enforce physical realism, a detachment condition is applied:

$$F_N(t) = \max [M(g + \ddot{u}_Z(t)), 0] \quad (1.12)$$

When $F_N(t) = 0$, the block is in a state of free-fall. During this time, it registers as **Status 3 (Detached/Gapping)**, and the horizontal frictional resistance drops strictly to zero.

Horizontal Dynamics and Coulomb Friction

The horizontal motion of the slab in the East-West (x) and North-South (y) directions is governed by 2D Coulomb friction and integrated using the Central Difference Method.

At any time step t , the absolute inertial position of the block—assuming strictly zero horizontal force—is predicted based on its previous positions:

$$x_{\text{inertia}} = 2x_t - x_{t-\Delta t} \quad (1.13)$$

$$y_{\text{inertia}} = 2y_t - y_{t-\Delta t} \quad (1.14)$$

To determine whether the block remains stuck to the ground or begins to slide, the model calculates the “trial” force required to force the block to move exactly with the ground. If the block remains stuck, its new absolute position must equal its previous position plus the incremental ground displacement ($\Delta u_x, \Delta u_y$):

$$x_{\text{stuck}} = x_t + \Delta u_x(t) \quad (1.15)$$

$$y_{\text{stuck}} = y_t + \Delta u_y(t) \quad (1.16)$$

According to Newton’s second law, the trial force required to maintain this stuck condition is:

$$F_{\text{trial},x} = \frac{M}{\Delta t^2} (x_{\text{stuck}} - x_{\text{inertia}}) \quad (1.17)$$

$$F_{\text{trial},y} = \frac{M}{\Delta t^2} (y_{\text{stuck}} - y_{\text{inertia}}) \quad (1.18)$$

The magnitude of this required driving force is $F_{\text{trial}} = \sqrt{F_{\text{trial},x}^2 + F_{\text{trial},y}^2}$. This is compared against the available dynamic friction capacity of the system, defined as $F_{\text{cap}} = \mu F_N(t)$.

Condition 1: Sticking (Status 1)

If $F_{\text{trial}} \leq F_{\text{cap}}$, the ground friction is sufficient to hold the block in place. The block moves exactly with the ground, maintaining **Status 1 (Stuck)**. The updated absolute positions become:

$$x_{t+\Delta t} = x_{\text{stuck}} \quad (1.19)$$

$$y_{t+\Delta t} = y_{\text{stuck}} \quad (1.20)$$

Condition 2: Sliding (Status 2)

If $F_{\text{trial}} > F_{\text{cap}}$, the inertial forces overcome the frictional resistance. The block slides, registering as **Status 2 (Slipping)**. The friction force acts at its maximum capacity (F_{cap}), directed exactly opposite to the trial direction of relative motion. The applied friction forces are isolated as:

$$F_{fx} = F_{\text{cap}} \frac{F_{\text{trial},x}}{F_{\text{trial}}} \quad (1.21)$$

$$F_{fy} = F_{\text{cap}} \frac{F_{\text{trial},y}}{F_{\text{trial}}} \quad (1.22)$$

The true absolute position of the sliding block is then updated using these bounded friction forces:

$$x_{t+\Delta t} = x_{\text{inertia}} + \frac{\Delta t^2}{M} F_{fx} \quad (1.23)$$

$$y_{t+\Delta t} = y_{\text{inertia}} + \frac{\Delta t^2}{M} F_{fy} \quad (1.24)$$

Boundary Collisions (Status 4)

To replicate physical constraints (e.g., headstone blocking the movement), a mathematical boundary condition is enforced. The wall is defined as a line in the horizontal plane:

$$y = x \tan(\theta) + b \quad (1.25)$$

where θ is the wall azimuth and b is the offset. After updating $x_{t+\Delta t}$ and $y_{t+\Delta t}$, the algorithm checks if the coordinate violates the boundary plane (e.g., crossing from the free side to the prohibited side). If a violation is detected, an inelastic collision is assumed. The block's position is geometrically projected exactly onto the wall's surface, and the velocities orthogonal to the wall are nullified, confining subsequent sliding strictly along the boundary or away from it. While pressed against this boundary, the block registers as **Status 4 (On Wall)**.

An Example of Modeled Slab Kinematics

Figure 1.38 illustrates an example of slab motion in response to a simulated acceleration seismogram, assuming a static friction coefficient of $\mu = 0.2$. The top row displays the input horizontal and vertical accelerograms, where the horizontal components have been rotated to be parallel and normal to the boundary wall. The shaded regions in the first row indicate the time intervals when the inertial force from the ground acceleration overcomes the static friction (sliding condition). During these periods, the slab detaches from the ground and begins to move relative to the wall or the ground. The second row displays the displacement when the wall blocks movement toward the NW, while the third row shows the scenario where the wall blocks movement toward the SE. The leftmost panels in the second and third rows depict the relative movement trajectory of the slab, colored by time to show the chronological evolution of the slip.

Figure 1.39 provides a more technical breakdown of the forward modeling kinematics for the NW slab movement scenario, i.e. the wall is defined to block the SE movement. The first

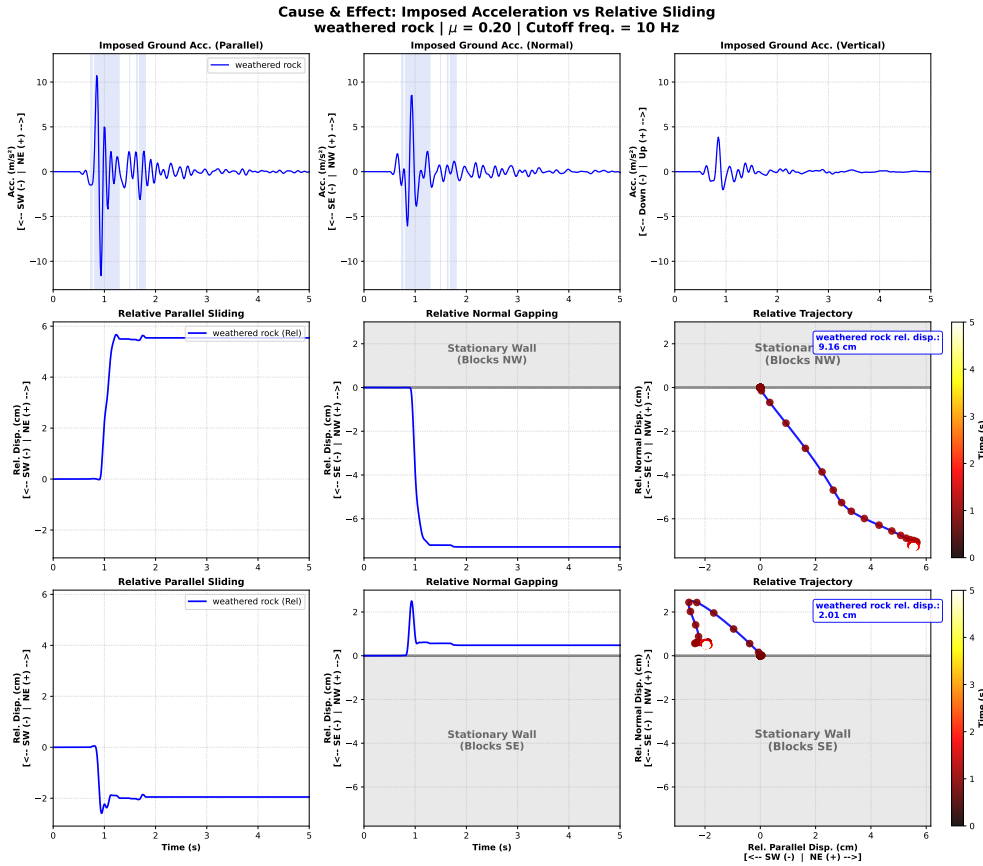


Fig. 1.38 Example of simulated slab motion driven by an input ground shaking described in a 3D seismogram, using a fixed friction coefficient of $\mu = 0.2$. The shaded regions in the top row indicate periods of active sliding.

row presents the absolute and relative movement trajectories, where the red line indicate the wall position. The right panel of the second row compares the required driving force (F_{trial} , representing the resultant inertial drive) against the available dynamic friction capacity of the system (F_{cap}). The left panel in the second row plots the cumulative slip (relative slab displacement) as a function of time, demonstrating that relative sliding occurs exclusively when $F_{\text{trial}} > F_{\text{cap}}$. Finally, the third row displays the three-component input accelerogram (left) alongside the kinematic status flag (ISTATUS) over time (right). As defined in the forward modeling methodology, the four physical states clearly correspond to the physical dynamics occurring at that exact moment.

Figure 1.40 follows an identical layout to Figure 1.39, but illustrates the kinematics for SE movement, where the boundary wall position has been reversed.

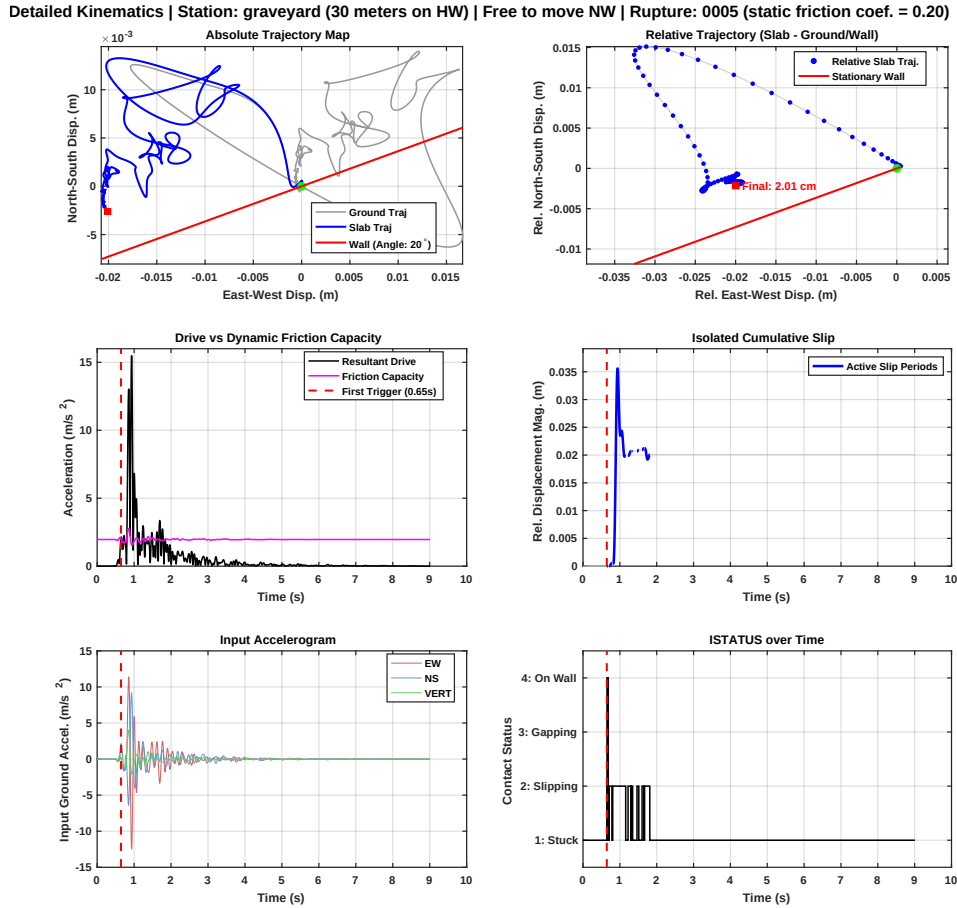


Fig. 1.39 Detailed kinematic breakdown of the forward model for a slab blocked by a NW wall, assuming a fixed friction coefficient of $\mu = 0.2$. The panels explicitly track the balance between inertial drive and frictional capacity, resulting in the corresponding contact status over time.

1.5.2 Influence of the Friction Coefficient on the SE-to-NW Displacement Ratio

Before performing the Bayesian update, I evaluated the impact of several key input parameters on the forward modeling of the slab displacement.

First, I investigated whether the choice of the static friction coefficient (μ) influences the ratio of SE to NW slab displacement. Specifically, I wanted to test if the field observations could be used to constrain the friction coefficient based on the observed polarity of the movement. The field data shows a clear polarity: the average slab displacement is 2.2 cm toward the NW and 4.7 cm toward the SE, yielding an observed SE-to-NW displacement ratio of approximately 2.14.

Figure 1.41 displays the simulated slab displacement ratio (SE vs. NW) across approximately 900 ruptures where the modeled displacement exceeded 1 cm. The friction coefficient is randomly distributed from 0.1 to 0.4. The results demonstrate that the simulated slab displacement inherently reproduce this polarity—meaning the SE slab displacement is consistently larger than the NW displacement—across all tested friction coefficients.

Detailed Kinematics | Station: graveyard (30 meters on HW) | Free to move SE | Rupture: 0005 (static friction coef. = 0.20)

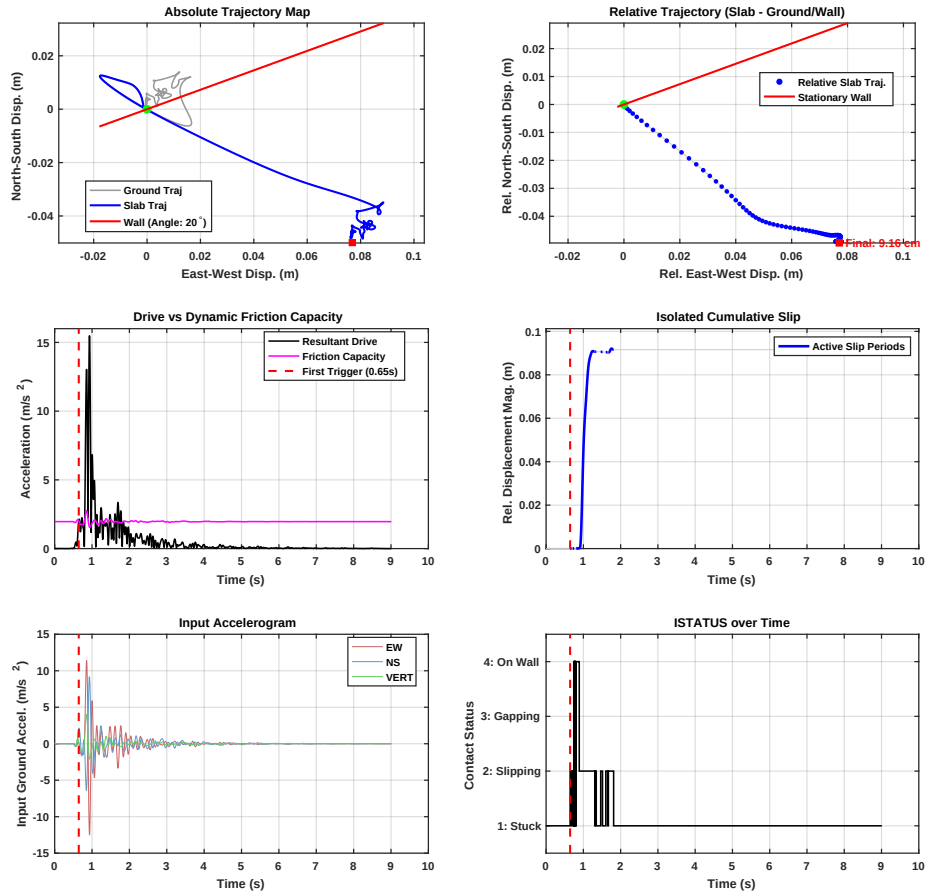


Fig. 1.40 Detailed kinematic breakdown of the forward model for a slab blocked by a SE wall, assuming a fixed friction coefficient of $\mu = 0.2$.

Because this polarity is driven by the asymmetrical nature of the ground motions rather than the specific friction parameter, the observed displacement ratio cannot be used to effectively constrain the static friction coefficient.

1.5.3 Influence of Cutoff Frequency on Slab Displacement Modeling

To determine the required frequency bandwidth of the simulated ground motions (accelerograms) for accurate slab displacement modeling, I repeated the forward modeling while varying the low-pass cutoff frequency of the input seismograms from 2 Hz to 10 Hz.

Figure 1.42 presents these modeling results, expressed as a consistency ratio: the slab displacement calculated at a given cutoff frequency divided by the baseline displacement calculated at 10 Hz. The top and bottom panels show the results for slab movement toward the SE and NW, respectively. The overall mean ratios across 1,000 ruptures are plotted as solid green and purple lines, while the results for the individual simulated ruptures are shown in gray. The results show that for SE movement, the consistency ratio reaches a plateau at a cutoff frequency of approximately 9 Hz, whereas for NW movement, the plateau is reached

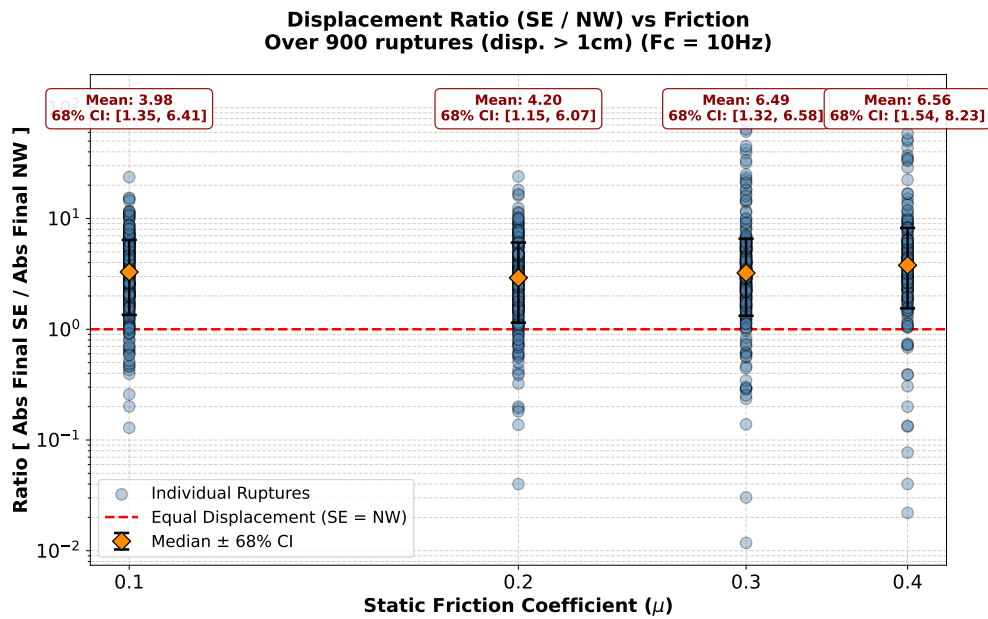


Fig. 1.41 Simulated slab displacement ratio between the SE and NW directions for ruptures generating at least 1 cm of slip. The friction coefficient is randomly chosen from 0.1, 0.2, 0.3 and 0.4.

at around 8 Hz. These findings indicate that a cutoff frequency of 10 Hz is sufficient to accurately model the slab displacement in both directions.

1.5.4 Influence of the Velocity Model on Slab Displacement Modeling

Finally, I investigated how the input velocity model impacts the slab displacement modeling. Figure 1.43 provides a demonstration of the same rupture simulated on both hard rock and weathered rock. In the top row, the orange line represents the seismogram simulated on hard rock, while the blue line represents the seismogram simulated on weathered rock. The second and third rows display the relative slab displacement parallel and normal to the wall and its 2D trajectory, respectively. The overall layout of this figure is identical to that of Figure 1.38.

Figure 1.44 quantifies this impact across the simulation catalog by displaying the ratio of outcomes between the weathered rock and hard rock models across 990 ruptures. On average, the inclusion of the weathered rock layer amplifies the input ground motion—specifically increasing the horizontal PGA by a factor of 1.54. This amplification, in turn, increases the modeled slab displacement by a factor of 1.86 for SE movement and 1.53 for NW movement, compared to the ground motions simulated on the hard rock.

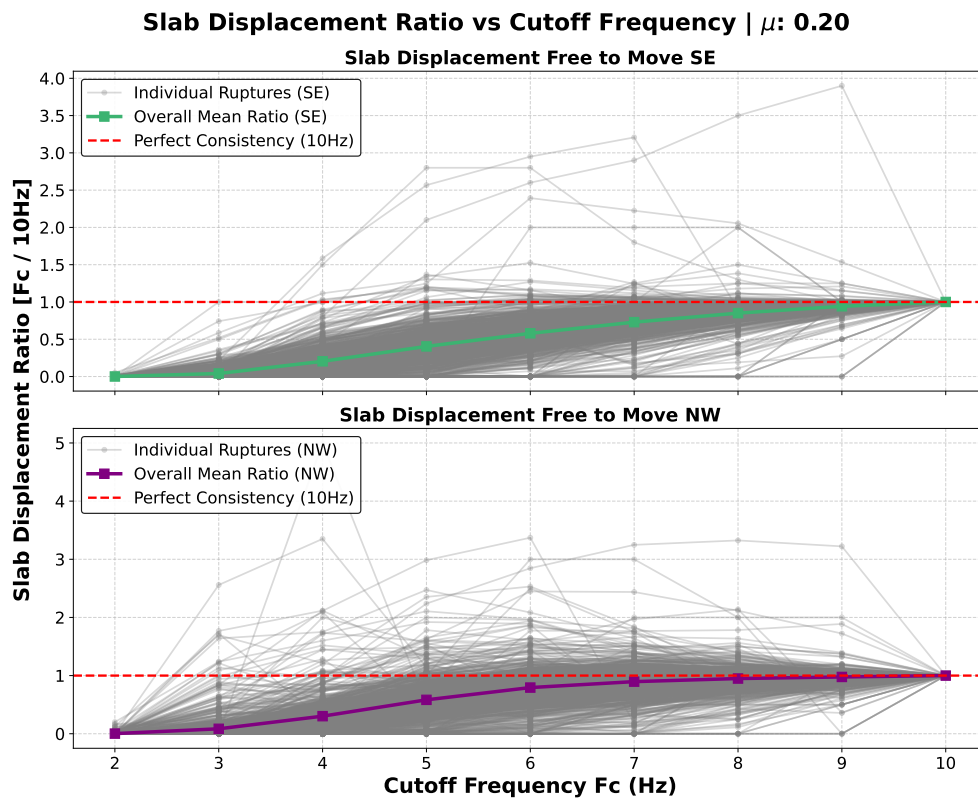


Fig. 1.42 Consistency ratio of the modeled slab displacement as a function of the low-pass cutoff frequency, normalized by the displacement calculated at 10 Hz. The top panel shows the ratios for slab movement toward the SE, and the bottom panel shows the movement toward the NW. The static friction coefficient is fixed at $\mu = 0.2$ for all runs.

1.5.5 Bayesian Update on Prior Horizontal PGA at Graveyard and Input Source

First, I investigated the correlation between the input source parameters and the resulting ground motion (using the horizontal PGA as a proxy for the intensity of the input seismogram).

Figure 1.45 displays the correlation matrix between five input source parameters and the horizontal PGA. The mean rise time is positively correlated with the weak zone taper strength. This relationship is expected, as a stronger taper effectively increases the mean rise time within the weak zone. Furthermore, a higher mean rise time lowers the peak slip rate across the fault, which consequently reduces the resulting horizontal PGA. This physical chain of effects explains why the PGA is negatively correlated with both the mean rise time and the weak zone taper strength.

Next, I evaluated the correlation between the four sets of slab observations—the slab displacements toward the SE and NW, and the observed azimuths toward the SE and NW — and the input static friction coefficient.

Figure 1.46 presents this correlation matrix. The simulated slab displacements in both the NW and SE directions are highly correlated with the horizontal PGA, and negatively

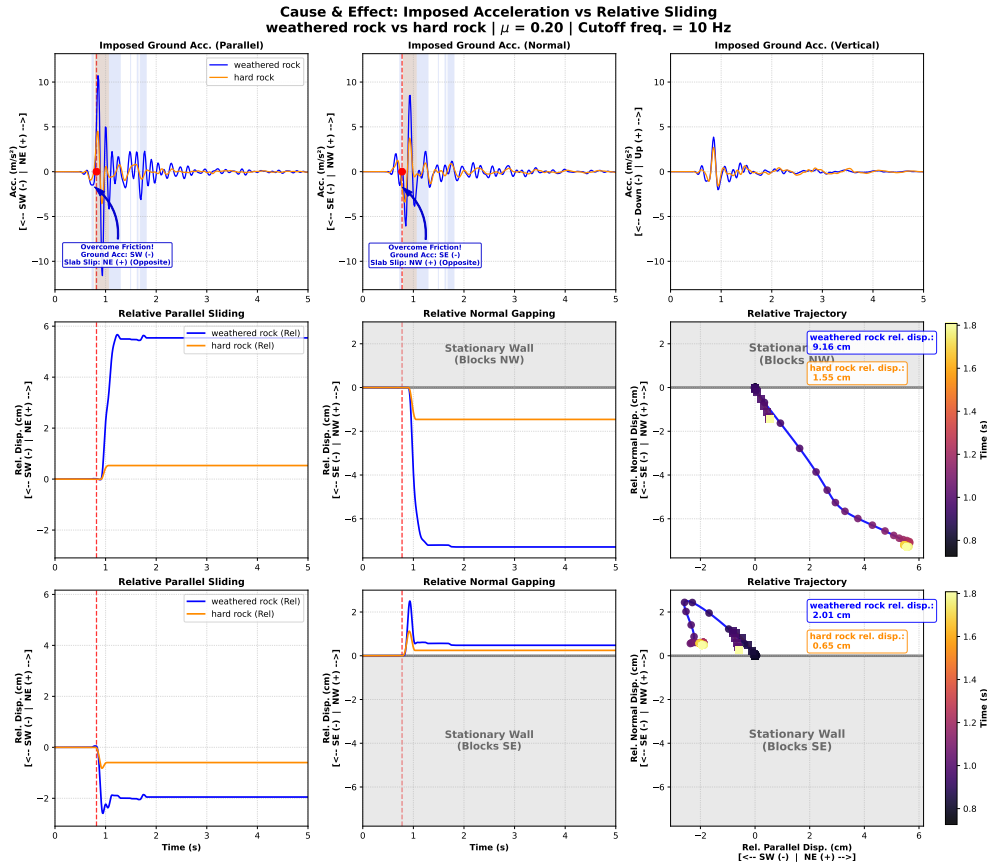


Fig. 1.43 An example of an imposed seismogram and two sets of modeled slab displacements evaluated on weathered and hard rock. Depending on the slab's initial position relative to the headstone, it is forced to move either NW or SE. A schematic configuration of the graveyard slabs is shown in the top right corner. The seismogram and corresponding slab motion are rotated into directions parallel and normal to the headstone. The leftmost panels show the relative displacement trajectories, with the color indicating the passage of time. In the top row (imposed horizontal acceleration), the shaded areas mark periods when inertial forces exceed friction, indicating when the slab actively slides.

correlated with the friction coefficient (μ). Moreover, the simulated displacement magnitudes in both directions are highly correlated with one another, meaning that greater slip in one direction strongly implies greater slip in the opposite direction. The displacement azimuths for both directions are also positively correlated with each other, though less strongly than the displacement magnitudes.

Similar to the methodology used in Figure 1.29, Figure 1.47a presents the diagnostic results for the four observational datasets: slab displacement to the NW (NWd), slab azimuth to the NW (NWa), slab displacement to the SE (SEd), and slab azimuth to the SE (SEa), evaluated as a function of the prior horizontal PGA. The red shaded zone indicates the standard deviation of the dataset. During the Bayesian update, a trust weight of 1.0 was assigned to the slab displacement measurements, while a reduced weight of 0.5 was assigned to the azimuths to reflect higher observational uncertainty. This reduced weight is justified because the overlap between the simulated and observed azimuth distributions is smaller than the overlap for the

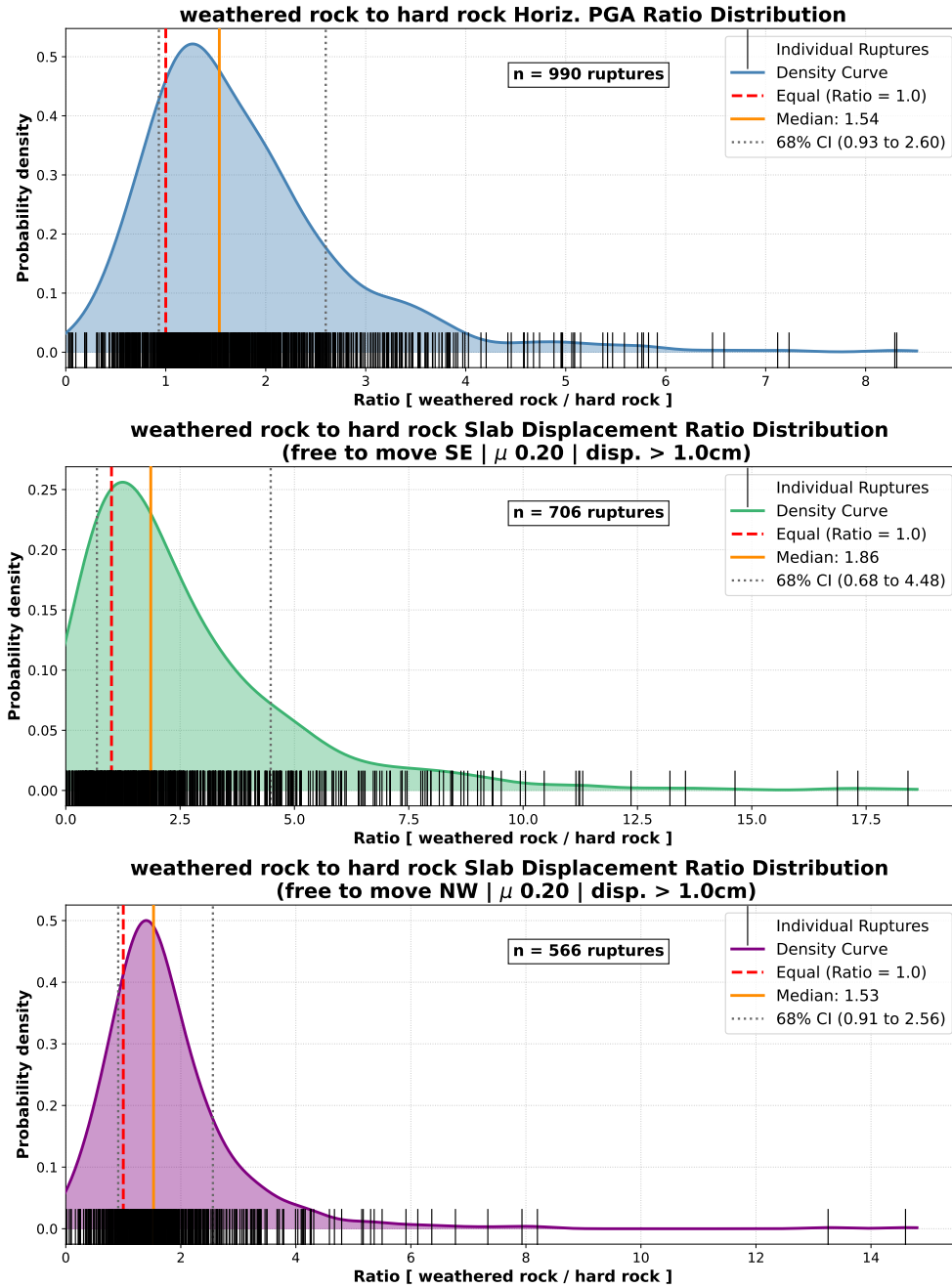


Fig. 1.44 Influence of the input velocity model on the modeled slab displacement. From top to bottom, the panels display the ratio (weathered rock / hard rock) of: the input horizontal PGA, the resulting slab displacement for SE movement, and the resulting slab displacement for NW movement. The low-pass cutoff frequency for the input seismograms is 10 Hz, and the static friction coefficient is fixed at $\mu = 0.2$.

displacement distributions in both directions, as previously demonstrated in Figures 1.36 and 1.37. therefore, I increase the *stdv* for azimuth to include more scenarios. Figure 1.47b illustrates the corresponding likelihood distributions for all individual observations.

Figure 1.48 illustrates the individual influence of the four datasets as well as their resulting joint likelihood. Ultimately, 62 out of the 997 evaluated ruptures yielded a joint relative likelihood greater than 1%. The maximum likelihood estimate (MLE, i.e., the highest peak)

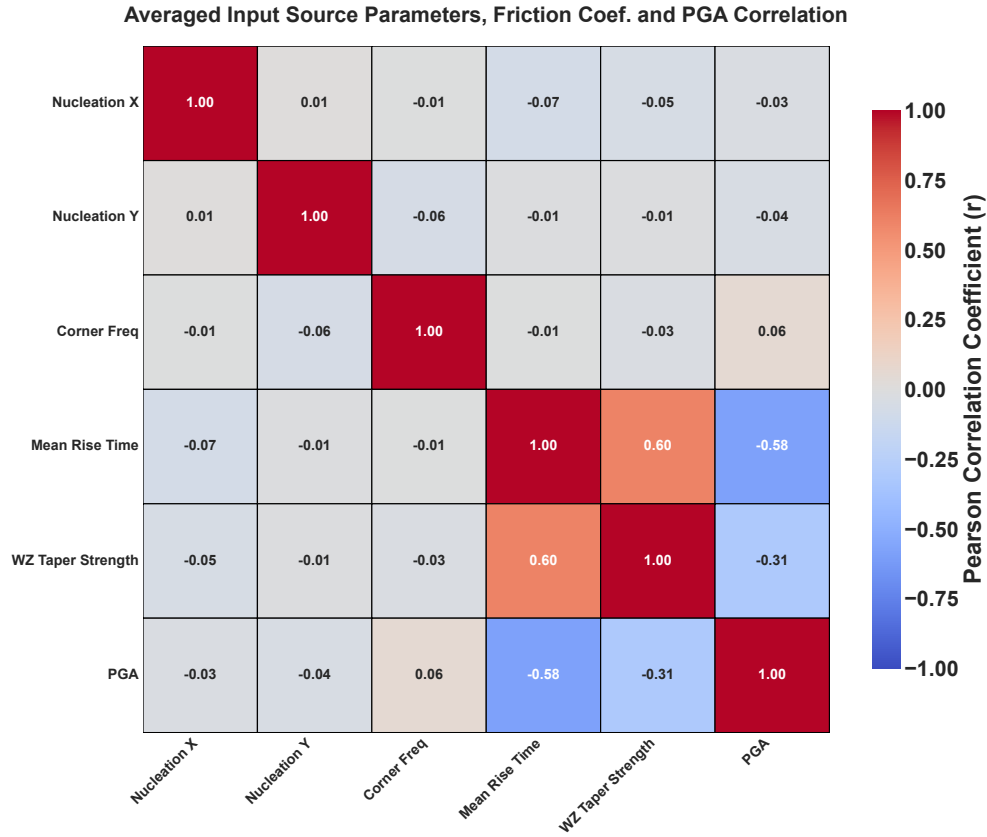


Fig. 1.45 Correlation matrix comparing five input source parameters with the resulting horizontal PGA.

and the median of the joint consensus both point to a horizontal PGA of 8.52 m/s^2 at the graveyard site. While this value is larger than the PGA estimated from the building damage, it is important to note that the input driving the forward slab modeling is the entire acceleration time series (seismogram), rather than just the isolated peak acceleration value.

Breaking down the individual datasets, 618 out of 997 ruptures achieved a $> 1\%$ relative likelihood for NWd. This was followed by 443 ruptures for NWa, 477 ruptures for SEd, and 893 ruptures for SEa. Consequently, only a small fraction of the simulated ruptures can simultaneously satisfy all four observational constraints.

Random Forest Sensitivity Analysis and Input Source Update

To evaluate whether the input seismic source parameters can be meaningfully constrained by the grave slab field observations, I performed a Random Forest sensitivity analysis on the four slab datasets and the horizontal PGA at the graveyard site.

The results reveal a clear separation between what controls the magnitude of the slab's movement versus its direction. The macroscopic source parameters (such as the mean rise time) dominate the predictions for displacement and shaking intensity. Specifically, the model achieves a high coefficient of determination (R^2) of 0.79 for both the NW displacement (NWd) and the horizontal PGA, and an R^2 of 0.66 for the SE displacement (SEd).

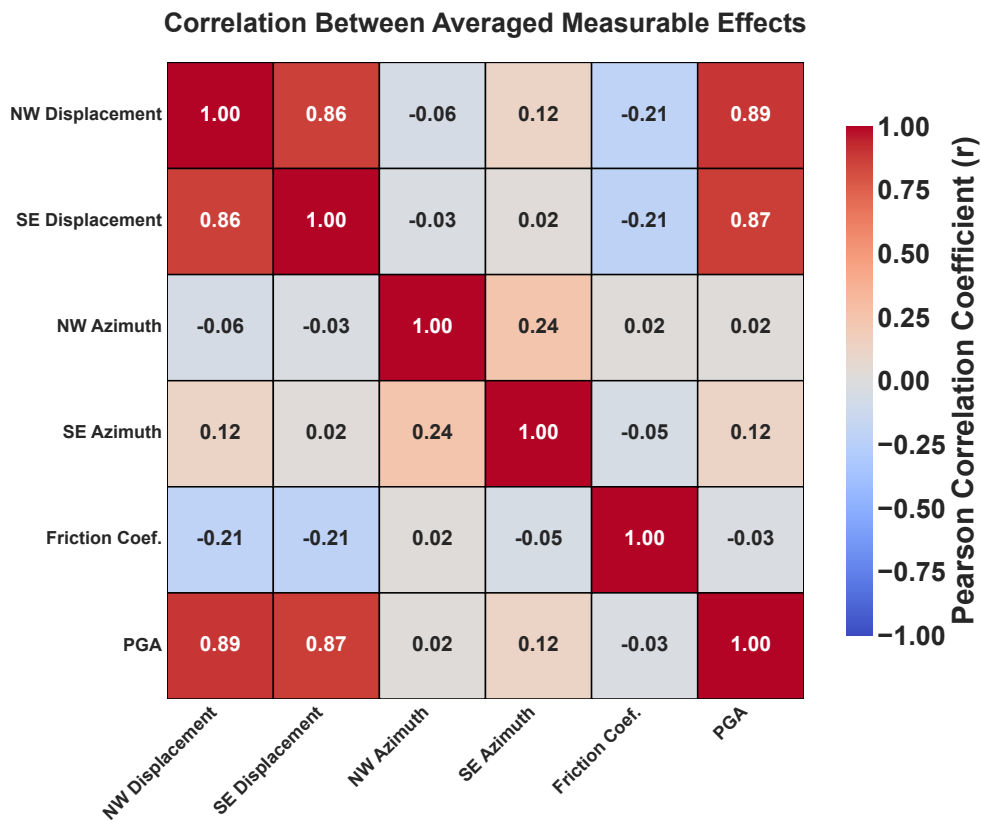
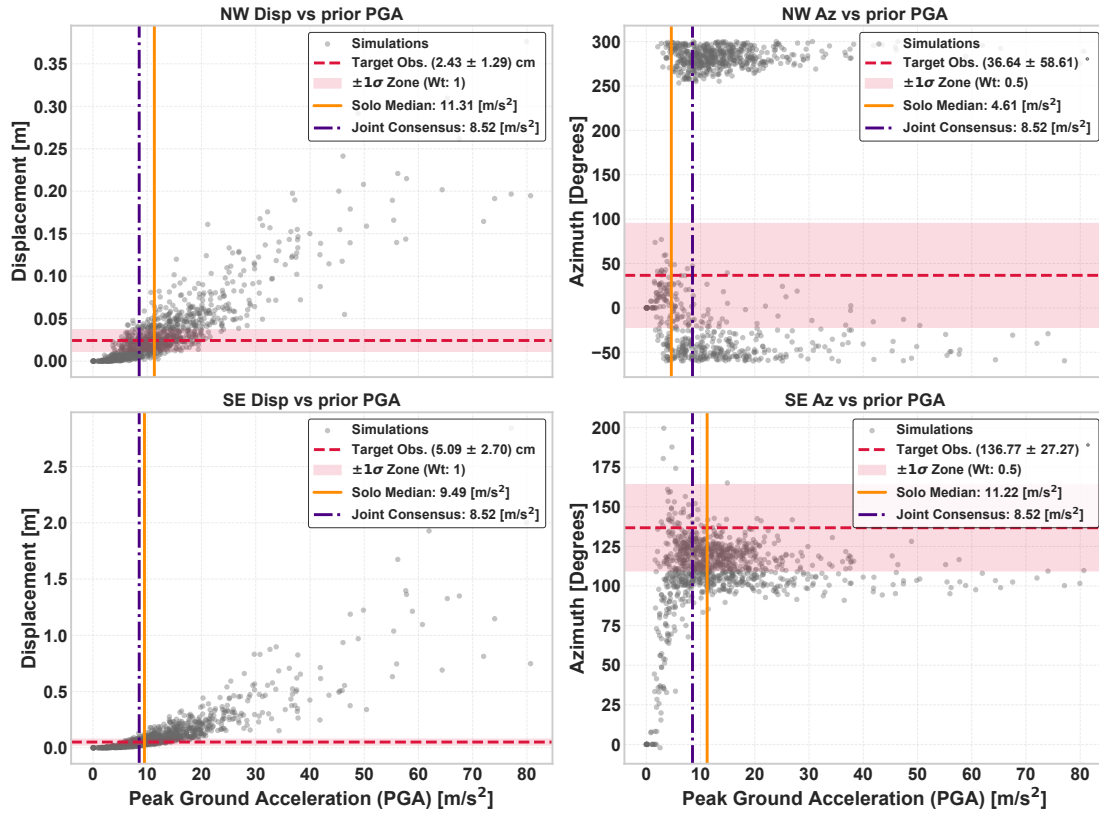


Fig. 1.46 Correlation matrix comparing the four modeled slab kinematics (displacement magnitude and azimuth for both NW and SE movements), the static friction coefficient (μ), and the horizontal PGA.

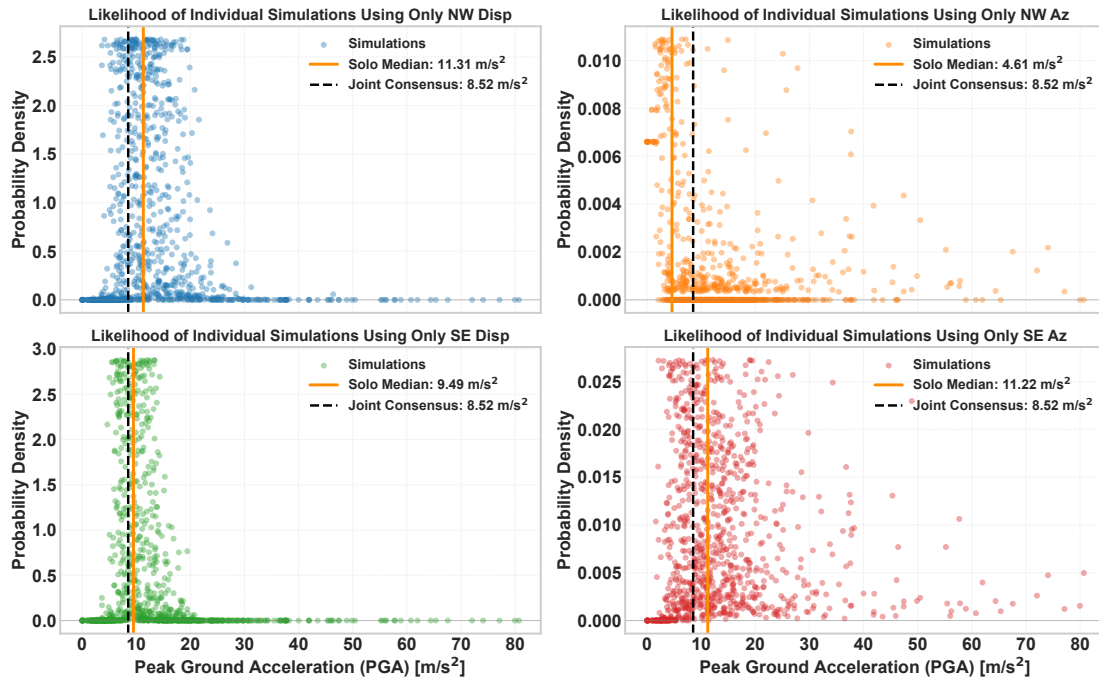
In contrast, the model is much less effective at predicting the sliding direction (azimuths), yielding significantly lower scores ($R^2 = 0.28$ for NWa and 0.46 for SEa). This indicates that the trajectory of the sliding slab is largely independent of the earthquake's source kinematics. Instead, the direction of movement could be more influenced by local, site-specific factors, such as topography and the geometric constraints of the retaining wall.

The Random Forest test demonstrates that some source parameters can be meaningfully constrained by the field observations, relying particularly on the slab displacement magnitude to constrain the mean rise time.

Figure 1.50 compares the updated (posterior) input source parameters for the slab modeling against their initial priors. Notably, while the overall posterior distribution for the mean rise time remains broadly similar to the prior, it now exhibits a much sharper peak around 0.4 s. Additionally, the posterior distribution for the friction coefficient shows a distinct preference for $\mu = 0.2$. Regarding the corner frequency, the highest observed peak corresponds to a mean rupture velocity of approximately 1500 m/s. As for the remaining input source parameters, the Random Forest analysis confirms that they cannot be meaningfully constrained by the grave slab data.



(a) Diagnostic results showing the posterior probability distributions for NWd, NWa, SEd, and SEa as a function of the prior PGA at the graveyard site.



(b) Likelihood weight distributions for each slab scenario derived from the Bayesian update.

Fig. 1.47 Comparison of slab diagnostic results and raw likelihood weight distributions for the Bayesian slab modeling analysis.

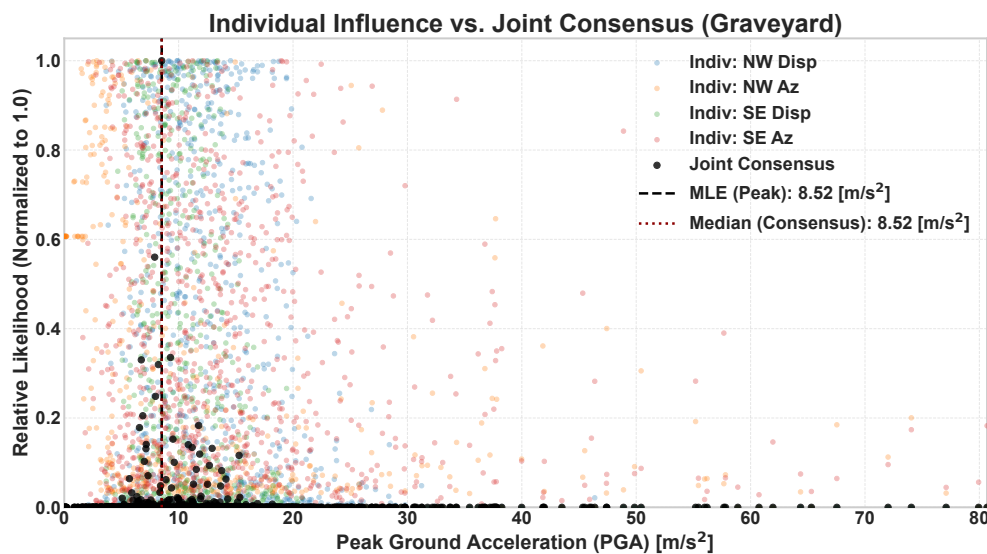


Fig. 1.48 Individual likelihood scatter plots for the four observational datasets and the resulting joint likelihood. The vertical dashed lines indicate the maximum likelihood estimate (MLE) and median values.

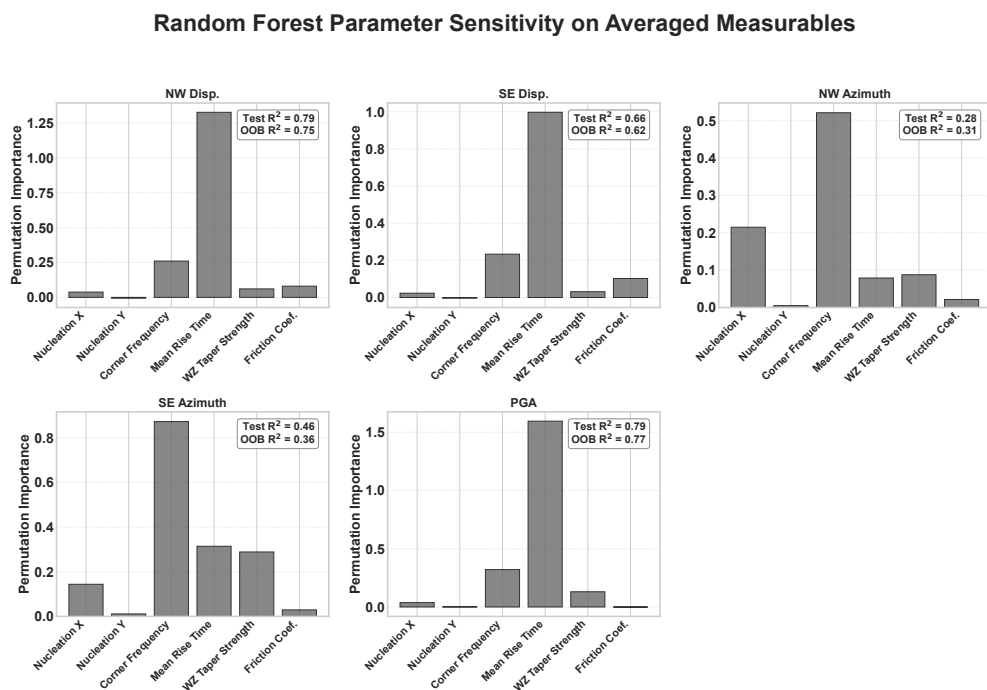


Fig. 1.49 Random Forest sensitivity analysis identifying parameter importance for the four slab measurable metrics and the horizontal PGA.

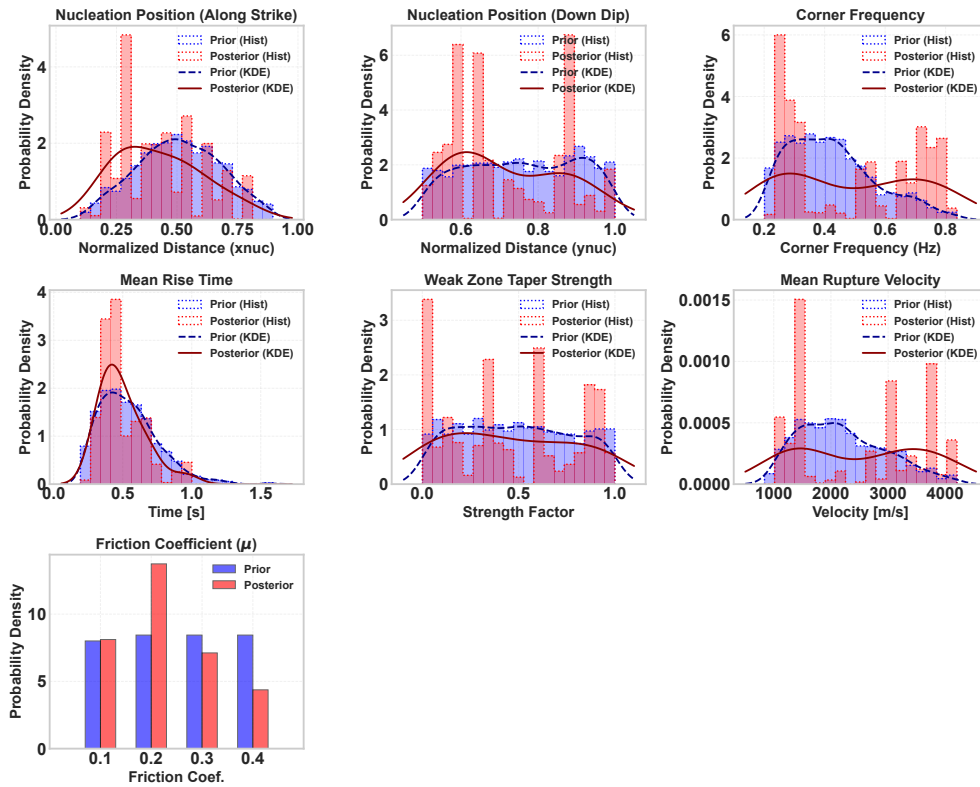


Fig. 1.50 Comparison of normalized prior and posterior distributions for the five input source parameters plus mean rupture velocity and friction coefficient using graveslab movement observations

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